

ANÉL HENNING | CYBER SECURITY | C. 813.743.1661 | ANEL_PURDUE@OUTLOOK.COM | COLLEGE: PURDUE GRAD ONLINE

Intro to Cybersecurity
Professor: Hany Othman

Statistics in IT
Professor: Jean Kotsiovos

Research Writing in IT
Professor: Jennifer Merritt

Foundation of Data Analysis
Professor: Saanie Sulley

Security for Analytics
Professor: Matthew Gonzalez

Cryptography Concepts Tech
Professor: Tiffany Laitola

Professor: Jean Kotsiovos

Intent: This paper explores purpose of Statistic Definition in Perspective in Sample Methods, between Discrete and Continuous Variables. And Descriptive Statistic, Correlation, Linear Regression, Probability, Random Variables and Distributions, Confidence Intervals, Hypothesis Testing, Chi-square, Inferential Statistics and ANOVA.

Programs: Microsoft Word and R-Studio



Statistic Definitions Perspective	2
Abstract	
This paper explores the purpose of Statistic Definitions in Perspective to prove your knowledge of statistics from a high-level point of view. Explain statistical terms, distinguish between a population and a sample. Describe the various sampling methods, distinguish between discrete and continuous variables with course outcome practiced in this assignment. Furthermore, I must explain statistical terms, concepts, and methods. Moreover, conduct research and use your book to explain the following statistical concepts: descriptive statistics, correlation and linear regression, probability, random variables and distributions, confidence intervals and hypothesis testing, Chi-square, inferential statistics, and ANOVA.	

Statistic Definitions Perspective	3
Descriptive statistics	
Descriptive statistics are numbers that describe data sets and key features which do not relate to one another, for example, someone's political standing. (Rumsey, 2019)	
The Mean is average of central tendency that calculates computer average for given set of numeric values (Chapple & NIJM, 2022). Median is central tendency to find midpoint value for all observations of a variable (Chapple & NIJM, 2022). Mode is the most frequent variable which occurs during observations, and it may not have a mode (Chapple & NIJM, 2022).	
These concepts introduce variance and interquartile data into the world of statistics where we can measure Frequence versus Relative Frequency. (Rumsey, 2019) We measure problems in middle of data percentage of data where first variable of height/weight second variable.	
Correlation and linear regression	
The correlation measures strength and direction of linear relationships such as x and y variables. The r is the correlation of the sample of data coefficient where we measure mean of x and y, and standard deviation. Furthermore, we measure Sample of x and the Sample of y plus the x mean and y mean. (Rumsey, 2019). Next, we measure how 'close' you get to negative one or one shows how strong linear relationships are. Here we use scatterplots with various correlations such as switching height and weight will not change the relationship such as formulas rise/run which is formula Y=MX+B we can use. Thus, we can accurately state if no data collected, we cannot measure interpolation. The problem we avoid is extrapolation: measuring the value of a variable based on relationship with another variable. Also, we do not make predictions if data is outside of the range of statistics (Rumsey, 2019, P.124). In scatterplots we can minimize influence over other factories with well-controlled medical experiments where x changes to y therefore you cannot change back to x. We call this the cause-and-effect relationship (Rumsey, 2019, P.124).	

Statistic Definitions Perspective	3
Descriptive statistics	
Descriptive statistics are numbers that describe data sets and key features which do not relate to one another, for example, someone's political standing. (Rumsey, 2019)	
The Mean is average of central tendency that calculates computer average for given set of numeric values (Chapple & NIJM, 2022). Median is central tendency to find midpoint value for all observations of a variable (Chapple & NIJM, 2022). Mode is the most frequent variable which occurs during observations, and it may not have a mode (Chapple & NIJM, 2022).	
These concepts introduce variance and interquartile data into the world of statistics where we can measure Frequence versus Relative Frequency. (Rumsey, 2019) We measure problems in middle of data percentage of data where first variable of height/weight second variable.	
Correlation and linear regression	
The correlation measures strength and direction of linear relationships such as x and y variables. The r is the correlation of the sample of data coefficient where we measure mean of x and y, and standard deviation. Furthermore, we measure Sample of x and the Sample of y plus the x mean and y mean. (Rumsey, 2019). Next, we measure how 'close' you get to negative one or one shows how strong linear relationships are. Here we use scatterplots with various correlations such as switching height and weight will not change the relationship such as formulas rise/run which is formula Y=MX+B we can use. Thus, we can accurately state if no data collected, we cannot measure interpolation. The problem we avoid is extrapolation: measuring the value of a variable based on relationship with another variable. Also, we do not make predictions if data is outside of the range of statistics (Rumsey, 2019, P.124). In scatterplots we can minimize influence over other factories with well-controlled medical experiments where x changes to y therefore you cannot change back to x. We call this the cause-and-effect relationship (Rumsey, 2019, P.124).	

Statistic Definitions Perspective	4
Random variables and distributions	
Rumsey (2019) states Random Variables and Distributions are coin flips of 100 times where n=100 and p=0.50. Here we introduce concepts of Standard Deviation, Variable Binomial Distribution, or Variable Binomial Random Mean relationships. Methods such as intuitive relationships measure probability of success with a p relationship. Furthermore, we measure proportion (p), and failure (1-p) divided by n-trials. Thus, the formula reads np(1-p) (Rumsey, 2019).	
Confidence intervals and hypothesis testing	
A Hypothesis testing is confidence interval which we measure between two value's population means divided by proportions. Here we build, calculate, and interpret hypothesis testing (Rumsey, 2019). When we assess a claim in narrow or wide confidence intervals, we can calculate our results (Rumsey, 2019). This sentence usually starts: "For these results ...," and stands for percentage of all possible random sample size of n of population (Rumsey, 2019). Thus, decent-quality data corrects confidence intervals, but data cannot be biased, and we sample low-quality data (Rumsey, 2019). In addition, if we work with sampling error, we question wording and practical difficulties in conducting surveys. Here we can introduce errors or bias in finding public opinion polls (Rumsey, 2019).	
Chi-square	
The definition of Chi-Square is a relationship between two variables quantitative and qualitative. Furthermore, chi-square relationships include quantitative data that continues into correlation and regression (Rumsey, 2019). For example, study time and exam scores are categorical relationships. Another one is categorical (gender and political relationships) turns into chi-square relationships we can examine (Rumsey, 2019). In doing so, we use hypothesis null (H0) and independent variables which do not have a relationship. If the cell count is more than five or equals five where we have p-value and confidence level is 'pre-determined' we can safely state Chi-Square relationship (Rumsey, 2019). Thus,	

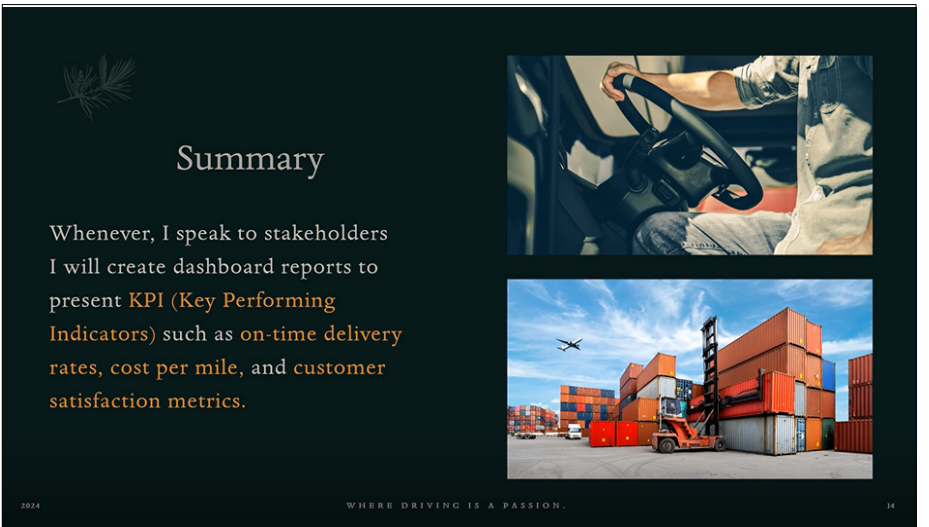
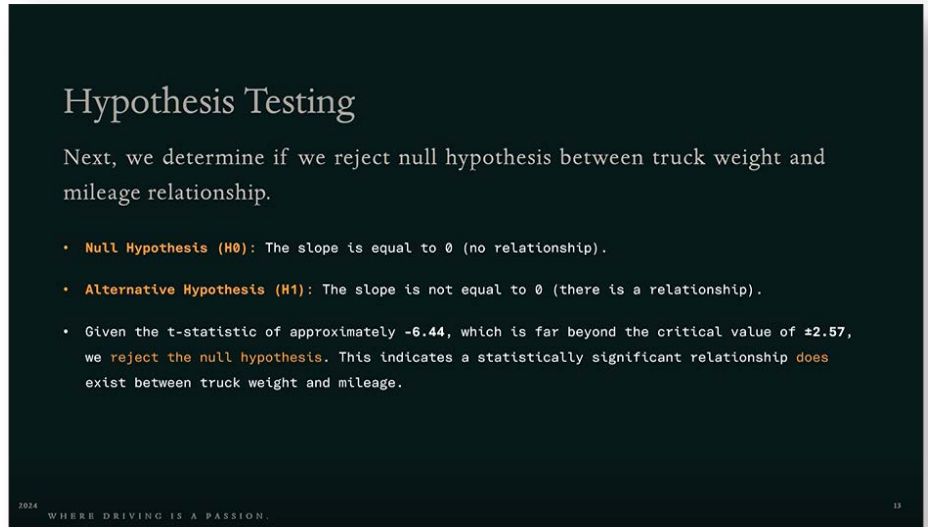
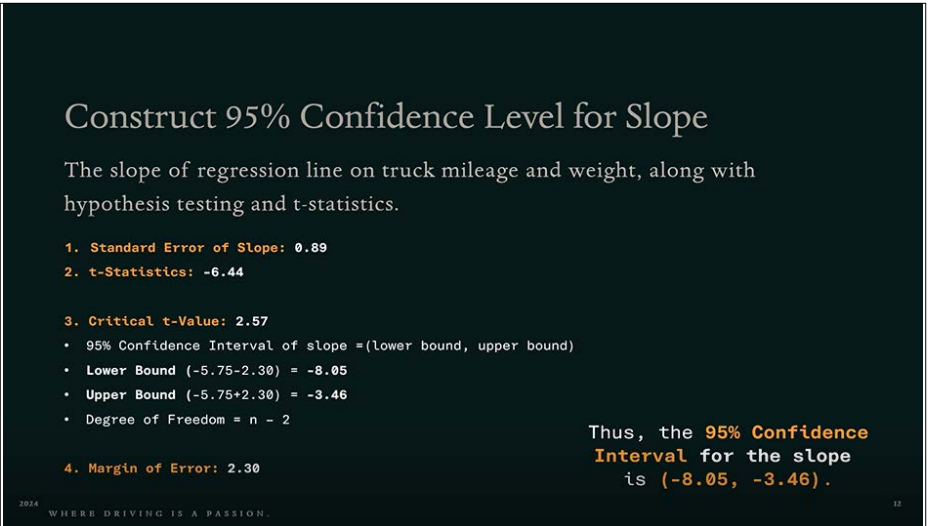
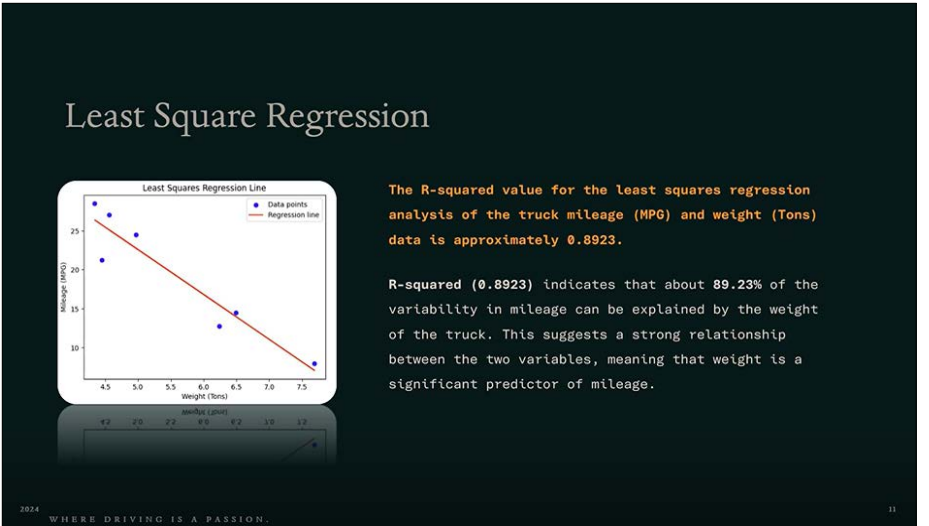
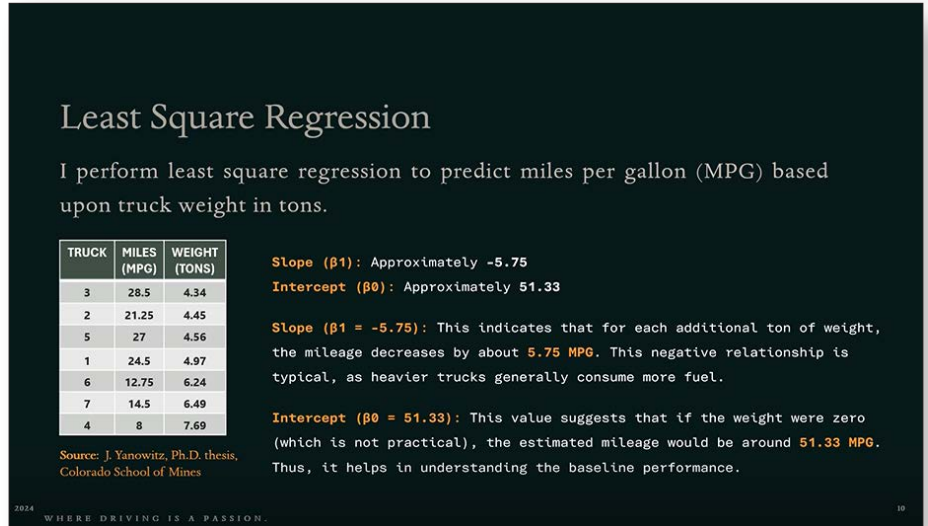
Statistic Definitions Perspective	5
our second Hypothesis (Ha) is the dependent variable. In conclusion, there is a significant difference between what you see and what we can expect (Rumsey, 2019). Furthermore, number of cell count is more than five or equal to five and we can combine cells to state in relationship (Rumsey, 2019).	
Inferential Statistics	
The definition of Inferential Statistics is a rest of data that calculate inferences or generalizations about characteristics of techniques (Bernstein, 1999). It is also known as a logical framework which creates inferential leaps of the sample population (Bernstein, 1999). Here we can measure probability theory and sample theory that goes into quality inferences. Because we want to see how close our probability relationship is to the truth (Bernstein, 1999).	
ANOVA Population	
The Anova Population is the balanced design where we measure the population means that are different to response of variables, y-character, and treatment variable of x-relationships (Rumsey, 2009). Thus, we design experiments with Random assigned to two relationships as treatment and responses compared for various treatment groups (Rumsey, 2009). Here, we compare regular populations as independent and K-populations as normal distribution (Rumsey, 2009). We use mu-formula of two variables and k-populations. Furthermore, we must use Hypothesis 2, F-test, and p-value of predetermined confidence level for further examination (Rumsey, 2009).	

Statistic Definitions Perspective	6
References	
Bernstein (1999). <i>Schaum's Outline of Theory and Problems of Elements of Statistics I</i> . Publisher: McGraw Hill.	
Chapple & NIJM (2022). <i>Data+ Study Guide, Exam DA0-001</i> . Publisher: Sybex. A Wiley Brand.	
Rumsey (2019). <i>Statistical Essentials for Dummies</i> Publisher: John Wiley & Sons, Inc.	
Rumsey (2009). <i>Statistical II for Dummies</i> Publisher: Wiley Publishing, Inc.	

Professor: Jean Kotsiosvos

Intent: Present to stakeholders for transportation company called TRUCKS® to discuss optimization of routes, monitor fuel efficiency, regular mainteance, leverage of data analytics and impvement of load management.

Programs: Microsoft PowerPoint and Microsoft Excel



References

Gendreau, M., & Potvin, J.-Y. (2010). *Handbook of metaheuristics* (2nd ed.). Springer.

Greene, d. L., & Plotkin, E. (2011). *Reducing greenhouse gas emissions from U.S. Transportation*. Transportation research board.

Upper, j. (2023). *Route analysis: A complete guide to techniques & benefits*. Upper. Retrieved from <https://www.Upperinc.Com/blog/route-analysis/>

Torres Arpi Acero, A., & González gil, F. (2021). *Fuel efficiency and safety in Coca-Cola FEMSA last-mile logistics*. Mit. Retrieved from <https://dspace.mit.edu/>

Shmueli, G., & Koppius, O. (2011). Predictive analytics in information systems research. *MIS quarterly*, 35(3), 553-572. <https://doi.Org/>

Moubray, J. (1997). *Reliability-centered maintenance*. Industrial press

Sivinski, J. (2012). The role of maintenance in the management of physical assets. *Journal of quality in maintenance engineering*, 18(1), 5-20. <https://doi.Org/>

Professor: Jean Kotsiovos
Intent: This paper explores purpops of Statistic Definition in Perspective to prove your knowledge of statistics from high-level point of view.
Programs: Microsoft Word



Statistic Definitions Perspective	2
Abstract	
This paper explores the purpose of Statistic Definitions in Perspective to prove your knowledge of statistics from a high-level point of view. Explain statistical terms, distinguish between a population and a sample. Describe the various sampling methods, distinguish between discrete and continuous variables with course outcome practiced in this assignment. Furthermore, I must explain statistical terms, concepts, and methods. Moreover, conduct research and use your book to explain the following statistical concepts: descriptive statistics, correlation and linear regression, probability, random variables and distributions, confidence intervals and hypothesis testing, Chi-square, inferential statistics, and ANOVA.	

Statistic Definitions Perspective	3
Descriptive statistics	
Descriptive statistics are numbers that describe data sets and key features which do not relate to one another, for example, someone's political standing. (Rumsey, 2019)	
The Mean is average of central tendency that calculates computer average for given set of numeric values (Chapple & NIJM, 2022). Median is central tendency to find midpoint value for all observations of a variable (Chapple & NIJM, 2022). Mode is the most frequent variable which occurs during observations, and it may not have a mode (Chapple & NIJM, 2022).	
These concepts introduce variance and interquartile data into the world of statistics where we can measure Frequence versus Relative Frequency. (Rumsey, 2019) We measure problems in middle of data percentage of data where first variable of height/weight second variable.	
Correlation and linear regression	
The correlation measures strength and direction of linear relationships such as x and y variables. The r is the correlation of the sample of data coefficient where we measure mean of x and y, and standard deviation. Furthermore, we measure Sample of x and the Sample of y plus the x mean and y mean. (Rumsey, 2019). Next, we measure how 'close' you get to negative one or one shows how strong linear relationships are. Here we use scatterplots with various correlations such as switching height and weight will not change the relationship such as formulas rise/run which is formula Y=MX+B we can use. Thus, we can accurately state if no data collected, we cannot measure interpolation. The problem we avoid is extrapolation: measuring the value of a variable based on relationship with another variable. Also, we do not make predictions if data is outside of the range of statistics (Rumsey, 2019, P.124). In scatterplots we can minimize influence over other factories with well-controlled medical experiments where x changes to y therefore you cannot change back to x. We call this the cause-and-effect relationship (Rumsey, 2019, P.124).	

Statistic Definitions Perspective	3
Descriptive statistics	
Descriptive statistics are numbers that describe data sets and key features which do not relate to one another, for example, someone's political standing. (Rumsey, 2019)	
The Mean is average of central tendency that calculates computer average for given set of numeric values (Chapple & NIJM, 2022). Median is central tendency to find midpoint value for all observations of a variable (Chapple & NIJM, 2022). Mode is the most frequent variable which occurs during observations, and it may not have a mode (Chapple & NIJM, 2022).	
These concepts introduce variance and interquartile data into the world of statistics where we can measure Frequence versus Relative Frequency. (Rumsey, 2019) We measure problems in middle of data percentage of data where first variable of height/weight second variable.	
Correlation and linear regression	
The correlation measures strength and direction of linear relationships such as x and y variables. The r is the correlation of the sample of data coefficient where we measure mean of x and y, and standard deviation. Furthermore, we measure Sample of x and the Sample of y plus the x mean and y mean. (Rumsey, 2019). Next, we measure how 'close' you get to negative one or one shows how strong linear relationships are. Here we use scatterplots with various correlations such as switching height and weight will not change the relationship such as formulas rise/run which is formula Y=MX+B we can use. Thus, we can accurately state if no data collected, we cannot measure interpolation. The problem we avoid is extrapolation: measuring the value of a variable based on relationship with another variable. Also, we do not make predictions if data is outside of the range of statistics (Rumsey, 2019, P.124). In scatterplots we can minimize influence over other factories with well-controlled medical experiments where x changes to y therefore you cannot change back to x. We call this the cause-and-effect relationship (Rumsey, 2019, P.124).	

Statistic Definitions Perspective	4
Random variables and distributions	
Rumsey (2019) states Random Variables and Distributions are coin flips of 100 times where n=100 and p=0.50. Here we introduce concepts of Standard Deviation, Variable Binomial Distribution, or Variable Binomial Random Mean relationships. Methods such as intuitive relationships measure probability of success with a p relationship. Furthermore, we measure proportion (p), and failure (1-p) divided by n-trials. Thus, the formula reads np(1-p) (Rumsey, 2019).	
Confidence intervals and hypothesis testing	
A Hypothesis testing is confidence interval which we measure between two value's population means divided by proportions. Here we build, calculate, and interpret hypothesis testing (Rumsey, 2019). When we assess a claim in narrow or wide confidence intervals, we can calculate our results (Rumsey, 2019). This sentence usually starts: "For these results ..." and stands for percentage of all possible random sample size of n of population (Rumsey, 2019). Thus, decent-quality data corrects confidence intervals, but data cannot be biased, and we sample low-quality data (Rumsey, 2019). In addition, if we work with sampling error, we question wording and practical difficulties in conducting surveys. Here we can introduce errors or bias in finding public opinion polls (Rumsey, 2019).	
Chi-square	
The definition of Chi-Square is a relationship between two variables quantitative and qualitative. Furthermore, chi-square relationships include quantitative data that continues into correlation and regression (Rumsey, 2019). For example, study time and exam scores are categorical relationships. Another one is categorical (gender and political relationships) turns into chi-square relationships we can examine (Rumsey, 2019). In doing so, we use hypothesis null (H0) and independent variables which do not have a relationship. If the cell count is more than five or equals five where we have p-value and confidence level is 'pre-determined' we can safely state Chi-Square relationship (Rumsey, 2019). Thus,	

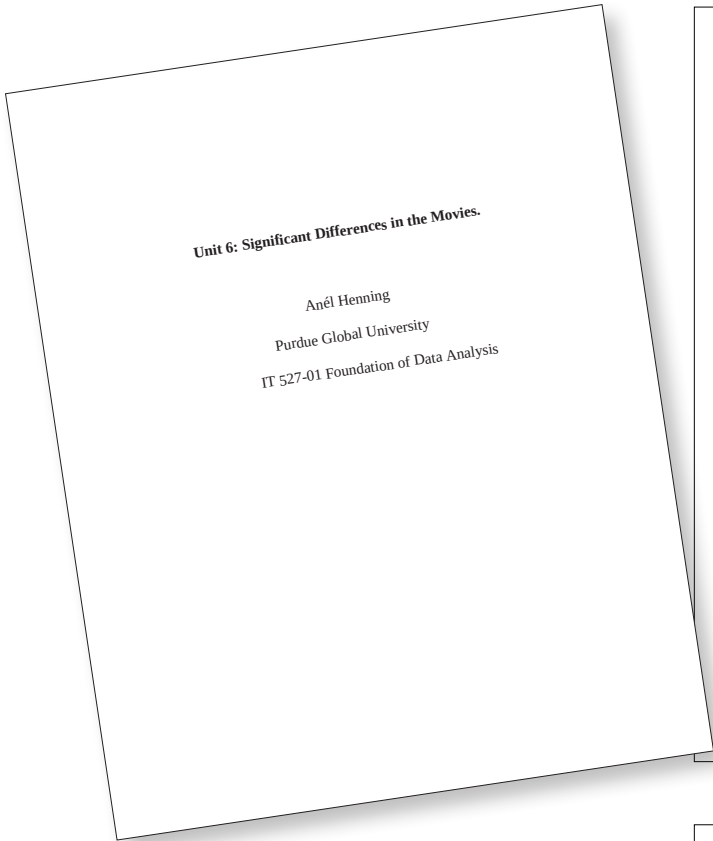
Statistic Definitions Perspective	5
our second Hypothesis (Ha) is the dependent variable. In conclusion, there is a significant difference between what you see and what we can expect (Rumsey, 2019). Furthermore, number of cell count is more than five or equal to five and we can combine cells to state in relationship (Rumsey, 2019).	
Inferential Statistics	
The definition of Inferential Statistics is a rest of data that calculate inferences or generalizations about characteristics of techniques (Bernstein, 1999). It is also known as a logical framework which creates inferential leaps of the sample population (Bernstein, 1999). Here we can measure probability theory and sample theory that goes into quality inferences. Because we want to see how close our probability relationship is to the truth (Bernstein, 1999).	
ANOVA Population	
The Anova Population is the balanced design where we measure the population means that are different to response of variables, y-character, and treatment variable of x-relationships (Rumsey, 2009). Thus, we design experiments with Random assigned to two relationships as treatment and responses compared for various treatment groups (Rumsey, 2009). Here, we compare regular populations as independent and K-populations as normal distribution (Rumsey, 2009). We use mu-formula of two variables and k-populations. Furthermore, we must use Hypothesis 2, F-test, and p-value of predetermined confidence level for further examination (Rumsey, 2009).	

Statistic Definitions Perspective	6
References	
Bernstein (1999). <i>Schaum's Outline of Theory and Problems of Elements of Statistics I</i> . Publisher: McGraw Hill.	
Chapple & NIJM (2022). <i>Data+ Study Guide, Exam DA0-001</i> . Publisher: Sybex. A Wiley Brand.	
Rumsey (2019). <i>Statistical Essentials for Dummies</i> Publisher: John Wiley & Sons, Inc.	
Rumsey (2009). <i>Statistical II for Dummies</i> Publisher: Wiley Publishing, Inc.	

Professor: Saanie Sulley

Intent: Prepare dataset for analysis by formatting, augmenting or reduce transformed variables and observations. An one-way ANOVA and explain Box Office Revenue by Significant Differences at the Movies.

Programs: Microsoft Word and R-Studio



Significant Differences at the Movies.

2

Abstract

Describe the Significant Differences at Movies meanwhile you must prepare a dataset for analysis by formatting, augmenting or reducing, and transforming variables and observations. The assignment will offer you the chance to practice your skill at deciding the statistical differences between groups in data. You will use the ANOVA techniques described in the text and in Seminar to examine the impact of certain variables in a data set on other variables.

Significant Differences at the Movies.

3

Quality Datasets used in Investigating Business Problems.

Prepare a dataset for analysis by formatting, augmenting or reducing, and transforming variables and observations.

- Download the Movies comma separated values (CSV) file. Import this data set into a data frame called Movies in RStudio. Place a screenshot of the data successfully loaded in RStudio into a Word document and label it appropriately.

Rank	Title	Studio	MPAA Rating	Worldwide Box Office	Domestic Box Office
1	Star Wars: The Last Jedi	SW	PG-13	693,995,000	654,000,000
2	Ready, Set, Go!	SW	PG	134,000,000	104,000,000
3	The Fate of the Furious	UM	PG-13	1,516,000,000	374,000,000
4	Deadpool 2	UM	PG	785,400,000	286,000,000
5	Avengers: Infinity War	MCU	PG-13	678,815,000	678,815,000
6	Spider-Man: Homecoming	MCU	PG-13	675,336,000	354,000,000
7	Star Wars: The Force Awakens	SW	PG	942,247,000	216,000,000
8	Guardians of the Galaxy Vol. 2	SW	PG-13	661,376,000	389,000,000
9	Thor: Ragnarok	MCU	PG-13	669,400,000	210,000,000
10	Wonder Woman	DC	PG-13	821,875,000	412,000,000
11	Dead	SW	PG	807,000,000	290,000,000
12	Phantom of the Caribbean: Dead Men Tell No Tales	SW	PG-13	764,000,000	170,000,000
13	Star Wars: The Rise of Skywalker	SW	R	947,400,000	247,000,000
14	Justice League	DC	PG-13	651,900,000	220,000,000
15	Logan	FX	R	419,000,000	230,000,000
16	Transformers: The Last Knight	FX	PG-13	654,000,000	130,000,000
17	Along with the Gods: The Two Worlds	MCU	PG-13	567,000,000	100,000,000
18	The Boss Baby	FX	PG	1,200,000,000	170,000,000
19	Deadpool	MCU	PG-13	782,900,000	360,000,000
20	Star Wars: The Rise of Skywalker	SW	PG-13	947,400,000	247,000,000
21	The Greatest Showman	FX	PG	419,000,000	170,000,000
22	Kingman: The Golden Circle	FX	R	419,000,000	100,000,000
23	The Mummy (2017)	UM	PG-13	412,000,000	90,000,000
24	Car 5	SW	G	400,000,000	100,000,000
25	Ricky Shave: Star	UM	R	370,000,000	110,000,000
26	Star Wars	SW	PG	500,000,000	100,000,000
27	Wonder on the Ocean Express (2017)	SW	PG-13	370,000,000	100,000,000
28	Star Wars: The Force Awakens	SW	PG-13	942,247,000	216,000,000
29	The Great Wall	UM	PG-13	369,000,000	40,000,000
30	Revenge of the Sith	SW	R	633,000,000	240,000,000

Showing 1 to 31 of 100 entries, 8 total columns

Significant Differences at the Movies.

4

2. Conduct a one-way ANOVA to decide if a movie's MPAA Rating (e.g. G, PG, PG-13, etc.) has a statistically significant impact on the movie's worldwide box office revenue. Place a screenshot of your ANOVA results in your Word document and explain whether there are any evidence ratings impacts worldwide revenue, using specific evidence from ANOVA results.

Box Office Revenue by MPAA Rating

Significant Differences at the Movies.

5

3. Repeat your ANOVA to decide if your findings in Step 2 are consistent when only considering domestic box office revenue. Place a screenshot of your ANOVA results and explain any differences between worldwide and domestic revenues.

Worldwide Box Office Revenue by MPAA Rating

Domestic Box Office Revenue by MPAA Rating

Significant Differences at the Movies.

6

4. Suppose you are a movie producer, and you make a new movie that you want to distribute. You are approached by Warner Brothers® (WB) and Buena Vista® (BV). Both studios are interested in distributing your movie, and you want to pick the studio that is likely to earn you the most money. Conduct an ANOVA to figure out if picking one studio or the other will make a difference in terms of your worldwide revenue.

Box Office Revenue by Studio

Significant Differences at the Movies.

9

References

Schmuller, J. (2017) *Statistical Analysis with R for Dummies*.[®] Publisher: John Wiley & Sons, Inc.

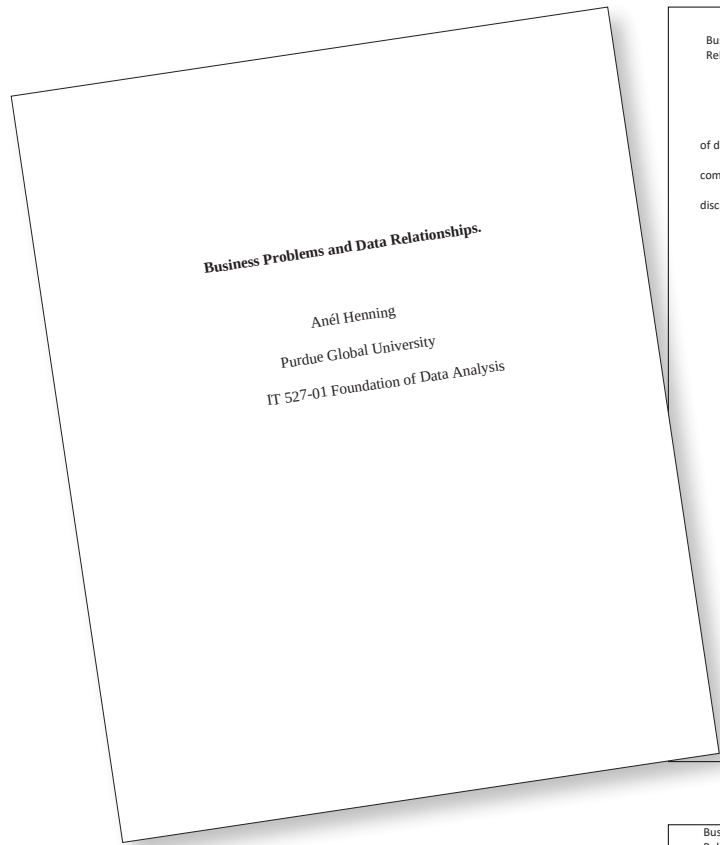
Rumsey (2009) *Statistical II for Dummies*.[®] Publisher: Wiley Publishing, Inc.

Bernstein (1999) *Schaum's Outline of Theory and Problems of Elements of Statistics I*.[®] Publisher: McGraw Hill.

Professor: Saanie Sulley

Intent: A paper explores purpose of business outline with document sources and types of data to measure business problems in R graphs and RStudio databases.

Programs: Microsoft Word and R-Studio



Business Problems and Data Relationships.

2

Abstract

This paper explores the purpose of an outline business problem to document sources and types of data needed to address an issue. In doing so, I would have to measure business problems and compare R graphs and RStudio datasets to do further analysis and read summary statistics. We start by discussing data governance in business practices.

Business Problems and Data Relationships.

4

R-Packages

When we installed "psych" package in R we can use procedures for psychological, psychometric and personality research. And when we generate descriptive statistics to describe what you have done with data as data analyst. Here is the screenshot of what we have done.

A box plot is a quick way to visualize data of horsepower versus number of cylinders in data frames. In analysis with box plot we review each box with dark black solid line inside box called median (Schmuller, 2017). The hinges are lower and upper edges of box with whiskers sticking out (Schmuller, 2017). We call this box-and-whiskers plot, and outside whiskers are outliers in lower and upper quartiles. Using the "plot" function we can demo simple scatterplot in R (Schmuller, 2017). Pick two numeric columns in your dataset and create a scatterplot showing their relationship (Schmuller, 2017). When we use a scatterplot in your Word document, we describe a scatterplot as useful to a data analyst (Schmuller, 2017). When we describe statistics in a data set, we use the function in "psych" library to generate descriptive statistics as 'describe' (Schmuller, 2017). Now we can create two numeric columns in your dataset and create a scatterplot showing their relationship (Schmuller, 2017). When we were done, we used common data formatting, transfer and manipulation issues and we can describe why such issues might be a problem for data analyst (Schmuller, 2017).

Business Problems and Data Relationships.

5

The psych package has been developed at Northwestern University to include functions most useful for personality and psychological research (Kotsiовs, 2024).

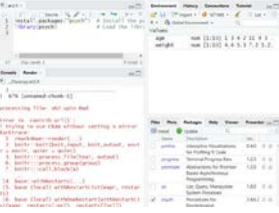


Figure 1: Loading psych packages

Business Problems and Data Relationships.

6

Describe statistics for dataset (screenshot)

When we review the screenshots of the graphs below, we can see how data analyst relates to it.

By starting with how to formulate questions, collect data to answer questions, describe data and draw conclusion with methods (Kotsiовs, 2024). We call these analytics using sampling, population vs sample, sample of convenience, types of data, design of experiments and bias in studios (Kotsiовs, 2024). When we draw Simple Scatterplot in R with "plot" functions we use concepts such as population and samples (Kotsiовs, 2024). We use samples of convenience not drawn by well-defined random methods. Meanwhile we have samples of convenience which may differ systematically from population. Lastly, we use stratified random sampling as population divided into groups called strata where simple random sample is stratum (Kotsiовs, 2024).

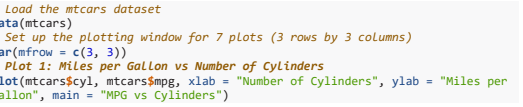


Figure 1: MPG vs Cylinders

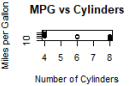
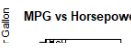


Figure 1: MPG vs Cylinders



Business Problems and Data Relationships.

7

Figure 2: MPG vs Horsepower

```
# Plot 3: Miles per Gallon vs Weight
plot(mtcars$wt, mtcars$mpg, xlab = "Weight", ylab = "Miles per Gallon", main = "MPG vs Weight")
```

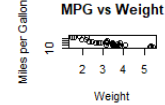


Figure 3: MPG vs Weight

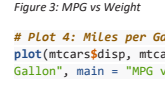
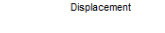


Figure 3: MPG vs Weight



Figure 4: MPG vs Displacement



Business Problems and Data Relationships.

9

References

Schmuller, J. (2017) *Statistical Analysis with R for Dummies*. Publisher: John Wiley & Sons, Inc.

Rumsey (2009) *Statistical II for Dummies*. Publisher: Wiley Publishing, Inc.

Bernstein (1999) *Schaum's Outline of Theory and Problems of Elements of Statistics I*. Publisher: McGraw Hill.

Professor: Saanie Sulley

Intent: A Power Testing Assignment conducts power analysis on Loan Applications data sets to decide sample size are sufficient including t-tests, ANOVA’s, and Correlation Analysis.

Programs: Microsoft Word and R-Studio



Power Testing Assignment

2

Abstract

This assignment aims to conduct power analyses on the Loan Applicants data set to decide if the sample sizes are sufficient for various statistical tests, including t-tests, ANOVAs, and correlation analyses. Power analysis helps find the likelihood that a study will detect an effect when there is an effect to be detected.

Power Testing Assignment

3

Importing Data

To begin with, the Loan Applicants data set was imported into RStudio and saved as a data frame called Loans. Below is a screenshot proving the successful import of the data.

The screenshot shows the RStudio interface. The Environment pane on the right displays a data frame named "Loan_Applicants" with 11 variables: Applicant.ID, Number.of.Missed.Late.Payments, Lines.of.Credit, C, R, Loan.Approval, age, weight, and others. The Console pane at the bottom shows the command "data <- read.csv('Loan_Applicants.R', as.is=TRUE)" and the output "data".

Power Testing Assignment

4

The R code used for the analysis is as follows:

- library(pwr)
- pwr.t.test(d = effect_size, sig.level = 0.05, power = 0.95, type = "two.sample")

Based on the results from Figure 2, the calculated sample size needed for 95% confidence was found to be sufficient. Consequently, the Loan Applicants data set has enough observations to conclude that Monthly Income influences loan approval decisions.

2. Power Analysis for ANOVA and Lines of Credit

Next, a power analysis was performed to decide if the Loan Applicants data set has sufficient samples to assess differences in loan applicants based on the number of lines of credit. The assumptions included an effect size of 0.3 and the following parameters:

- Alpha level: 0.05 (for 95% confidence).
- Power: 0.99.

The R code used for this analysis is:

- pwr.anova.test(k = number_of_groups, f = 0.3, sig.level = 0.05, power = 0.95)

The results in Figure 3 show that the sample size is adequate for detecting differences among groups based on lines of credit. Therefore, the Loan Applicants data set supports sufficient observations for ANOVA analysis.

3. Power Analysis for Correlation

Finally, to assess the strength of relationships between various attributes (excluding Applicant ID and Make Loan), a power analysis for correlation was performed with an assumed

Power Testing Assignment

5

The parameters used were:

- Alpha level: 0.05 (for 95% confidence).
- Power: 0.99.

The R code for this analysis is as follows:

pwr.r.test(r = 0.25, sig.level = 0.05, power = 0.95)

The analysis presented in Figure 4 suggests that the Loan Applicants data set has enough observations for performing correlation analysis, ensuring reliable results.

In conclusion, the power analyses conducted for the Loan Applicants data set show that there are sufficient sample sizes for t-tests, ANOVAs, and correlation analyses. Each analysis proved that the Loan Applicants data set can reliably assess the influence of Monthly Income on loan approval, differences in lines of credit, and relationships between various attributes.

	Applicant.ID
1	931569
2	736182
3	787832
4	858956
5	764782
6	918761

	Number.of.Missed.Late.Payments
1	12
2	12

Power Testing Assignment

8

The screenshot shows the RStudio interface. The Console pane at the bottom displays the output of the power analysis, including the calculated sample size (n = 398) for a two-sample t-test with an effect size of 0.25, alpha level of 0.05, and power of 0.95. The Environment pane on the right shows the data frame "data" with 13 variables: Applicant.ID, Number.of.Missed.Late.Payments, Lines.of.Credit, Credit.Score, Reliability, Monthly.Income, Age.at.First.Credit, Age.in.Years, Marital.Status, Make.Loan, and others.

Statistic Definitions Perspective

6

References

Bernstein (1999). *Schaum's Outline of Theory and Problems of Elements of Statistics I*. Publisher: McGraw Hill.

Chapple & NIJM (2022). *Data+ Study Guide, Exam DA0-001*. Publisher: Sybex. A Wiley Brand.

Rumsey (2019). *Statistical Essentials for Dummies* Publisher: John Wiley & Sons, Inc.

Rumsey (2009). *Statistical II for Dummies* Publisher: Wiley Publishing, Inc.

Professor: Saanie Sulley

Intent: Describe quality and format of datasets used in investigating business problems. The data use formulas, statistical methods and simulations used in data analysis.

Programs: Microsoft Word

Quality Datasets used in Investigating Business Problems.

Anél Henning

Purdue Global University

IT 527-01 Foundation of Data Analysis

Quality Datasets used to Investigate Business Problems.

2

Abstract

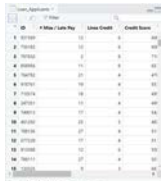
Describe the quality and formatting of datasets used in investigating business problems.

Associate a business problem with data that could be used to address it. Distinguish between data that is relevant to a business problem and data that is not. Explain formulas, statistical methods, and simulations used for business data analytics. Analytics begins with basic statistical analysis and the ability to visualize data in ways that answer questions. This Assignment will require you to implement several functions and techniques in RStudio to analyze and visualize data and interpret the results.

Quality Datasets used to Investigate Business Problems.

4

1. Download the [Loan Applicants](#) comma separated values (CSV) file. Import this data set into a data frame called Loans in RStudio. Place a screenshot of the data successfully loaded in RStudio into a Word document and label it appropriately.



2. Create histograms of the Number of Missed/Late Payments and Monthly Income attributes.

Place screenshots of these two histograms into your Word document, labeled appropriately.

Explain how a data analyst would interpret these histograms. What do they mean when you look at them? Refer to section 6.3 of the textbook if you need help.

The two histograms are labeled properly and explain how they would interpret them and here they refer to section 6.3 of textbook for further help.

Load necessary libraries

Load necessary library

library(ggplot2)

Quality Datasets used to Investigate Business Problems.

5

3. Create a boxplot of the reliability attribute. Place a screenshot of this boxplot in your Word document. Explain how a data analyst would interpret the boxplot.
 - a. What does it mean when you look at it?
 - b. Refer to section 6.5 of the textbook if you need help.

A boxplot of reliable structure interprets how to refer to section 6.5 and how the analysis should interpret boxplot.

Box Plot

ggplot(data = loan_data, aes(y = LoanAmount)) +

Plot the histogram

```
1 # Load necessary libraries
2 loan_data <- read.csv("loan_applications.csv")
3
4 # Display first few rows of dataset
5 head(loan_data)
6
7 # Load necessary library
8 library(ggplot2)
9
10 # Histogram
11 ggplot(data=loan_data, aes(x=LoanAmount)) +
12   geom_histogram(bins=width=500, fill="blue", col="black")
13
14 # Boxplot
15 ggplot(data=loan_data, aes(y=LoanAmount)) +
16   geom_boxplot()
17 # Plot histogram
18 hist(loan_data$LoanAmount, breaks=c(100, 200, 300, 400, 500, 600, 700, 800, 900, 1000))
```

(c) Analyze the Histogram

To decide whether histogram is unimodal or bimodal, visually inspect the histogram you created.

- Unimodal: The histogram has one clear peak.

- Bimodal: The histogram has two distinct peaks.

You can also use the following R code to analyze the histogram peaks more systematically, by

Quality Datasets used to Investigate Business Problems.

6

Example of checking the frequency values (you can adapt this as needed)

```
max_frequencies <- max(frequency_distribution$counts)
```

```
mode_count <- sum(frequency_distribution$counts == max_frequencies)
```

```
if (mode_count > 1) {
  cat("The histogram is bimodal.\n")
} else {
  cat("The histogram is unimodal.\n")
}
```

A standard Pearson correlation and its attributes expect Applicant ID and Make Loan is not needed and they use it for correlation matrix and explain how two sets of variables are strongly correlated in your explanation.

4. Create a standard Pearson correlation between all attributes except Applicant ID and Make Loan. Place a screenshot of your correlation matrix in your Word document and explain which

two sets of variables are the most strongly correlated and how you know. Be sure to use the correlation matrix results in your explanation.

Correlation

```
correlation <- cor(loan_data$AnnualIncome, loan_data$LoanAmount, use = "complete.obs")
print(correlation)
```

5. Run an independent t-test using the Make Loan attribute as the two-factor group attribute,

and Credit Score as the dependent attribute. Place a screenshot of your t-test results in your Word document and write an explanation of the results. What does the t-test result mean in terms of Credit Score's influence on Make Loan?

Quality Datasets used to Investigate Business Problems.

7

model function where they create summary functions with upper and lower estimate values. We use the 'confint' function in RStudio Program. Thereby, we can calculate the duration of confidence level from derived interval by setting lower and upper estimates.

T-test results write explanation of results in terms of Credit Score's Influence on Make Loan.

```
>plot(Loan$eruptions, Loan$waiting)
```

```
>lme <- lm(eruptions ~ waiting, data=Loan)
```

```
>summary(lme)
```

1. Frequency Distribution: Constructed using class width of 15 starting from a lower limit of 30.

2. Histogram: Created from the frequency distribution.

3. Unimodal or Bimodal: Determined based on the peaks observed in the histogram.



Quality Datasets used to Investigate Business Problems.

8

References

Schmuller, J. (2017) *Statistical Analysis with R for Dummies*. Publisher: John Wiley & Sons, Inc.

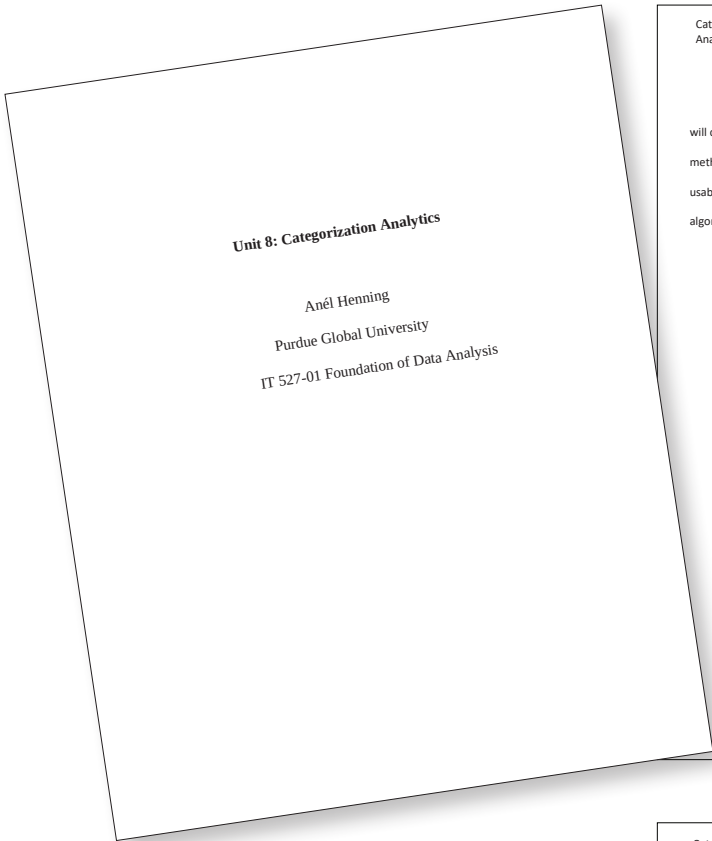
Rumsey (2009) *Statistical II for Dummies*. Publisher: Wiley Publishing, Inc.

Bernstein (1999) *Schaum's Outline of Theory and Problems of Elements of Statistics I*. Publisher: McGraw Hill.

Professor: Saanie Sulley

Intent: This assignment will enable you to practice Categorizational Analytics in RStudio. Especially, when they create and interpret k-means analyze categorizal data analysis.

Programs: Microsoft Word



Categorization Analytics.2

Abstract

This Assignment will enable you to practice categorization analytics in RStudio. Specifically, you will create and then interpret a k-Means model. Analyze categorical situations in data analysis, statistical methods to data sets to create categories and interpret the results of categorical models. Construct usable and effective data analytics models incorporating industry-recognized software and standard algorithms.

Categorization Analytics.3

Statistical Analysis Tools Comparison.

Statistical analyst for trucking company, TRUCKS, here I conduct research on two statistical analysis software that you can use with Excel, comparing them. Work with both numbers and labels of categories. Quantitative data consists of numerical values while qualitative data consists of text values which describe characteristics, traits and attitudes such as Pet Name, Animal Type and Breed Name (Chapple & NIJIM, 2024). Here I can ask questions such as Why? And What? (Chapple & NIJIM, 2024). By finding the structure of data types I can find types of organization necessary for different categories. For example: structured, unstructured or semi-structured categories (Chapple & NIJIM, 2024).

Age	Marital_Status	Gender	Weight_Category	Cholesterol	Stress_Management	Trait_Anxiety
25	0	0	0	180	0	0
45	1	1	1	220	1	1
30	2	0	2	250	0	0
35	3	1	0	190	1	1
50	1	1	1	210	0	0
40	0	0	2	240	1	1
60	2	1	1	230	0	0
55	3	0	0	200	1	1

Load necessary libraries

library(dplyr)

library(ggplot2)

Sample patient data

patients_df <- data.frame(
 Age = c(25, 45, 30, 35, 50, 40, 60, 55),
 Marital_Status = c(0, 1, 2, 3, 1, 0, 2, 3),

Categorization Analytics.5

Cluster == 2 ~ "Moderate Risk",
 Cluster == 3 ~ "High Risk",
 Cluster == 4 ~ "Critical Risk"
))

View the PatientClusters data frame
print("Patient Clusters DataFrame:")
print(PatientClusters)

Recommendation for Critical Risk patients
critical_risk_patients <- PatientClusters %>% filter(Risk_Category == "Critical Risk")
recommendation <- "Attend stress management classes and monitor cholesterol levels regularly."
print("Recommendation for Critical Risk Patients:")
print(recommendation)

Relationship between Gender and Risk Categories
gender_risk_relationship <- table(PatientClusters\$Gender, PatientClusters\$Risk_Category)
print("Relationship between Gender and Risk Categories:")
print(gender_risk_relationship)

Categorization Analytics.6

[1] PatientClusters: data frame:
> print(PatientClusters)
Age Marital_Status Gender Weight_Category Cholesterol Stress_Management
1 25 0 0 0 180 0
2 45 1 1 1 220 1
3 30 2 0 2 250 0
4 35 3 1 0 190 1
5 50 1 1 1 210 0
6 40 0 0 2 240 1
7 60 2 1 1 230 0
8 55 3 0 0 200 1
Trait_Anxiety Cluster Risk_Category
1 30 2 Moderate Risk
2 70 1 Low Risk
3 50 3 High Risk
4 40 2 Moderate Risk
5 60 1 Low Risk
6 80 4 Critical Risk
7 90 4 Critical Risk
8 20 2 Moderate Risk
>
> # Recommendation for Critical Risk patients
> critical_risk_patients <- PatientClusters %>% filter(Risk_Category == "Critical Risk")
> recommendation <- "Attend stress management classes and monitor cholesterol levels regularly."
>
> # Relationship between Gender and Risk Categories
> gender_risk_relationship <- table(PatientClusters\$Gender, PatientClusters\$Risk_Category)
> print("Relationship between Gender and Risk Categories:")
[1] "Relationship between Gender and Risk Categories:"
> print(gender_risk_relationship)
Critical Risk High Risk Low Risk Moderate Risk
0 1 1 0 2
1 1 0 2 1
>

Categorization Analytics.4

Weight_Category = c(0, 1, 2, 0, 1, 2, 1, 0),
Cholesterol = c(180, 220, 250, 190, 210, 240, 230, 200),
Stress_Management = c(0, 1, 0, 1, 0, 1, 0, 1),
Trait_Anxiety = c(30, 70, 50, 40, 60, 80, 90, 20)
)

Create k-means clustering model with four clusters
set.seed(123) # For reproducibility
kmeans_result <- kmeans(patients_df[, c(3)], centers = 4) # Exclude Gender for clustering

Add cluster assignments to the original data frame
patients_df\$Cluster <- kmeans_result\$Cluster

Create a centroid table
centroid_table <- as.data.frame(kmeans_result\$centers)
centroid_table\$Size <- table(patients_df\$Cluster)

Print centroid table
print("Centroid Table:")
print(centroid_table)

Create a new data frame called PatientClusters

Categorization Analytics.8

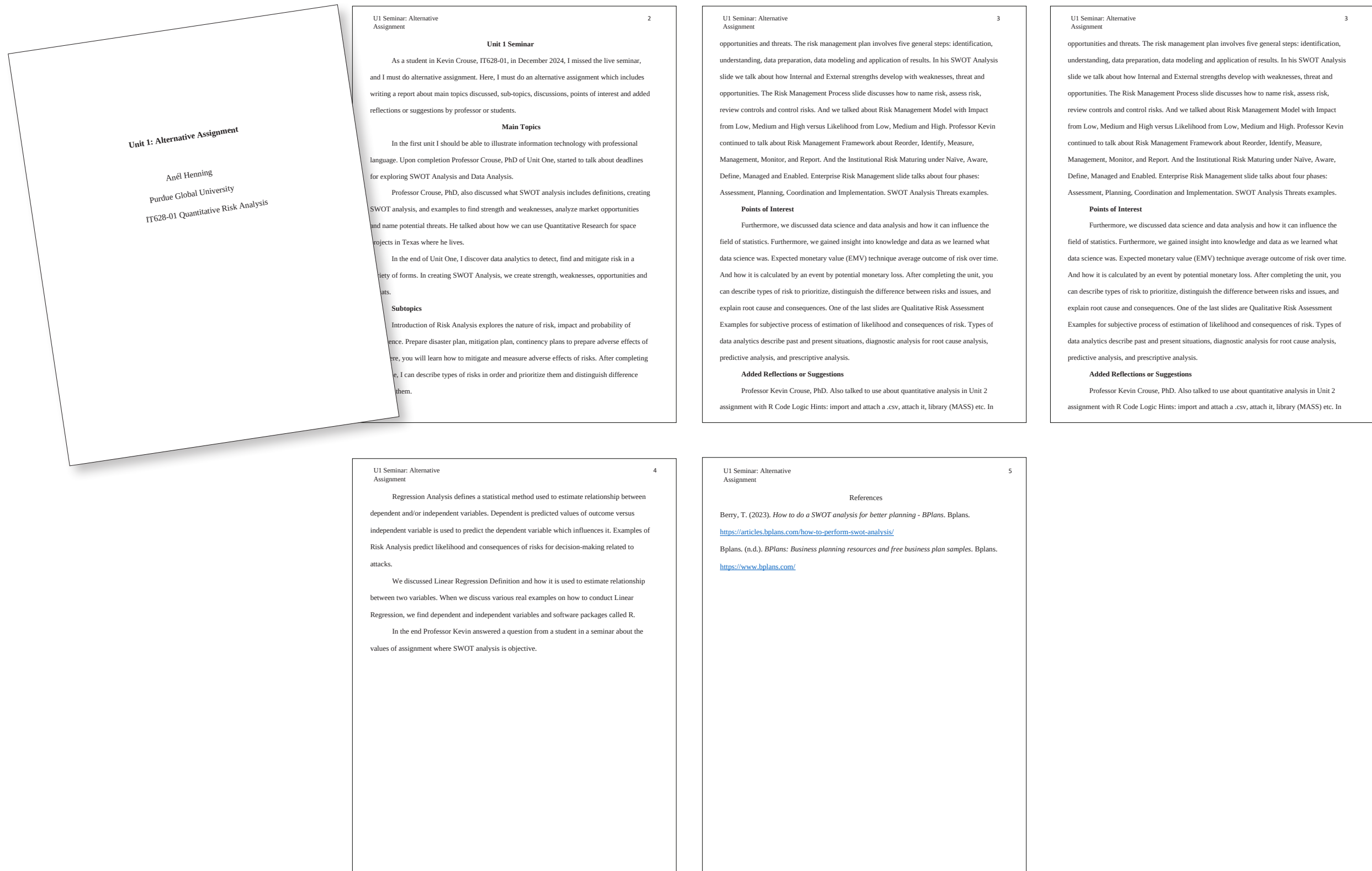
References

Gendreau, M., & Potvin, J.-Y. (2010). *Handbook of metaheuristics* (2nd ed.). Springer.
Greene, D. L., & Plotkin, E. (2011). *Reducing greenhouse gas emissions from U.S. transportation*. Transportation Research Board.
Upper, J. (2023). *Route analysis: A complete guide to techniques & benefits*. Upper. Retrieved from <https://www.upperinc.com/blog/route-analysis/>
Torres Arpi Acero, A., & González Gil, F. (2021). *Fuel efficiency and safety in Coca-Cola FEMSA last-mile logistics*. MIT. Retrieved from <https://dspace.mit.edu/bitstream/>
National Academies of Sciences, Engineering, and Medicine. (2020). *Reducing fuel consumption and greenhouse gas emissions of medium- and heavy-duty vehicles, phase two: Final report*. Washington, DC: The National Academies Press. doi:10.17226/25542. Retrieved from <https://nap.nationalacademies.org/read/25542/chapter/11>
Shmueli, G., & Koppius, O. (2011). Predictive analytics in information systems research. *MIS Quarterly*, 35(3), 553-572. <https://doi.org/10.2307/23042796>
Moubray, J. (1997). *Reliability-centered maintenance*. Industrial Press
Sivinski, J. (2012). The role of maintenance in the management of physical assets. *Journal of Quality in Maintenance Engineering*, 18(1), 5-20. <https://doi.org/10.1108/13552511211215745>
Coyle, J. J., Langley, C. J., & Gibson, B. J. (2016). *Supply chain management: A logistics perspective* (10th ed.). Cengage Learning.

Professor: Kevin Crouse

Intent: Research paper assignment about Risk Analysis and it's concepts such as SWOT analysis, Correlation and Linear Regression.

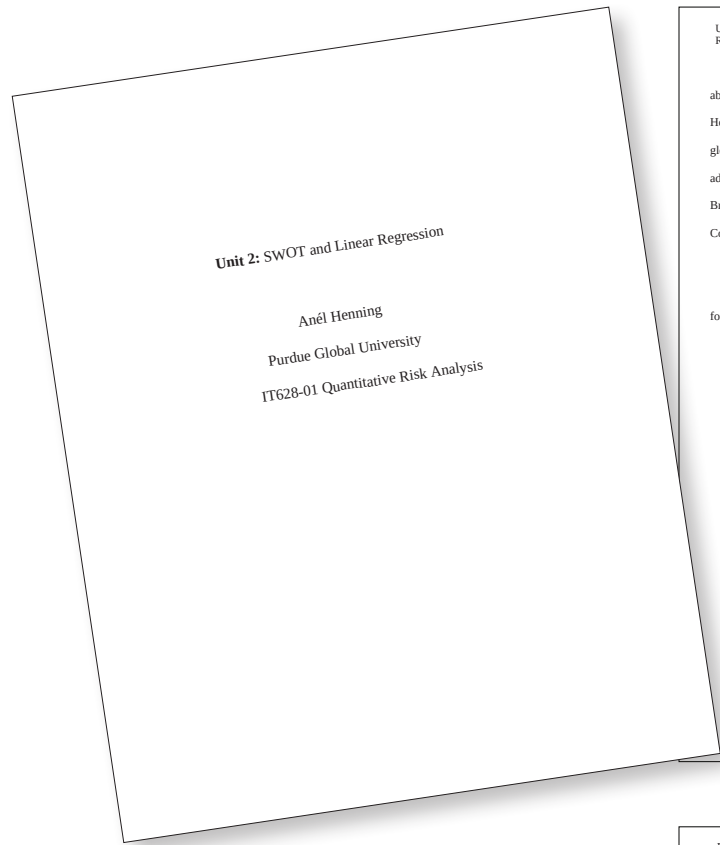
Programs: Microsoft Word



Professor: Kevin Crouse

Intent: This paper is about *Home & Heart’s Annual Report* by IncFact (2024) and their SWOT analysis using R for linear regression. A discussion of interpretation of model in R for insulation rating, outdoor temperature and number of occupants and their age.

Programs: Microsoft Word



Unit 2: SWOT and Linear Regression2

The Home & Hearth’s Annual Report is published by IncFact. (2024) and tells facts about as Home & Hearth’s size, annual revenues, industry and other names it runs in. Home and Heart are known for radiant heat gas fireplace logs, burner pans, grates, gas fittings, silica sand, glowing embers and highlight paint for logs sold as a unit (IncFact, 2024). Their competitive advantage comes from better offers in pricing of products and services such as Summit Pressed Brick & Tile, Belden Brick & Sales, or obtaining cheaper concrete such as Superior Ready-Mix Concrete, Ocean Glass tile, Diamond Concrete supply, etc. (IncFact, 2024).

SWOT Analysis Table

Using SWOT analysis helps find and plan for mitigation of risk and prepare model in R for linear regression. A SWOT analysis, linear regression and can model the assignment.

Strength • 40 years+ in market • Understand customer • Strong supplier in crude oil	Weaknesses • Oil turn rancid after few months • Risk of wastage • Struggles - Inventory Management • Lead to shortage or surpluses
Opportunities • Tech Advancement • Improve oil-burn technique • Market expand urban-to-rural	Threats • Price Volatility • Fluctuate oil price • Impact profit + plan • Access Natural Gas • Heat oil decrease

Figure 1: SWOT Analysis in Heating Oil

Unit 2: SWOT and Linear Regression3

Screenshot of Linear Regression Model

Interpretation of model

As they interpret models, they focus on coefficient and p-value of independent variables. Coefficients are relationships between predictors and response variables, for example: Heating_Oil_Used column name.

A positive coefficient for Temperature would suggest higher temperatures, for example:

Unit 2: SWOT and Linear Regression4

Furthermore, p-values decides the significance of each predictor less than 0.05 is statistically significant. Thereby, if p-value < 0.05 = contributes meaningfully to predict Heating_Oil_Used column name.

Call:

lm(formula = Heating_Oil_Used ~ Insulation_Rating + Outdoor_Temp + Num_Occupants + Home_Age + Home_Size, data = HeatingOil)

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	161.699	25.534	6.33	1.7e-09 ***
Insulation_Rating	3.331	0.563	5.91	4.7e-08 ***
Outdoor_Temp	-0.886	0.135	-6.57	4.3e-10 ***
Num_Occupants	-0.305	0.419	-0.73	0.465
Home_Age	2.002	0.258	7.75	3.3e-12 ***
Home_Size	0.0035	0.0005	6.98	1.8e-11 ***

Unit 2: SWOT and Linear Regression5

Summary of model

The linear regression model helps Home & Heart mitigate risk by predicting heat oil and how to manage inventory of company to reduce risk of shortages or surpluses. They decided Variable Analysis in R for Intercept, Insulation_Rating, Outdoor_Temp, Num_Occupants and Home_Age.

Intercept	161.699	Baseline heat oil is zero, depending on scale.
Insulation_Rating	3.331, p-value < 0.001	significant predictor p-value < 0.05.
Outdoor_Temp	-0.886 units	Every 1-degree increase in temperature decreases by 0.886 units, factors constant. Significant predictor.
Num_Occupants	-0.305, p-value > 0.05	Not significant, and insignificant effect on heating oil.
Home_Age	2.002, p-value < 0.001	Highly significant predictor, older homes consume more heating oil.
Home_Size	0.0035, p-value < 0.001	Significant prediction where larger homes consume more heating oil with an increase of 0.0035 units for each square unit home.

Risk Management

Significant Predictors: Insulation_Rating, Outdoor_Temp, Home_Age, Home_Size.

Not Significant: Num_Occupants

Support in research

In reading Home & Hearth’s annual report, they figured out Superior Ready-Mix Concrete is a risk because of the most revenue they obtain (IncFact, 2024).

Unit 2: SWOT and Linear Regression6

References

BPlans. (n.d.). Learn to build a better business plan. <https://www.bplans.com/>

Kida, Y. (2019, September 23). Generalized linear models - Towards Data Science. Medium

Towards Data Science. <https://towardsdatascience.com/generalized-linear-models-9cbf848bb8ab>

IncFact. (2024). *Home & Hearth Revenue, Growth & Competitor Profile*. Retrieved December 12, 2024, from <https://incfact.com/company/homehearth-sanmarcos-ca/>

Professor: Kevin Crouse

Intent: This research paper predicts Loan Default Risk using Generalized Linear Models (GLM) which focus on lenders face, loan default risk, sentiment analysis and risk management. Futhermore, is a Risk Management Plan with R

Programs: R-Studio

Unit 6: Predict Loan Default Risk using Generalized Linear Models (GLM)

Anél Henning
Purdue Global University
IT513 Research and Writing in IT

U4 Predict Loan Default Risk 2

Unit 4 GLM and Loan Default Risk

In utilizing Purdue Global's library, they able to find relevant articles which focus risk lenders face, loan default risk, sentiment analysis and risk management in lending. The authors in Global Banking News (GBN) wrote in 2023, about Loanspark, a German commercial lending service that offers specific needs to businesses. For example: partners or co-brands with other financial agencies operating savings banks and shift from decentral to central lending solutions. Here, they suggest credit rationing overlooks soft info that would live in decentral system. Hence, Loanspark produces an innovative process accessible by commercial clients which connects to financial institutions (Global Banking News, 2023).

Secondly, they found an article about decentralized lending with sentiment score analysis of peer-to-peer on Android applications in Indonesia (Kristanti and Ramantoko, 2023). Because of sentiment analysis they were able to evaluate user reviews and feedback of overall peer to peer lending landscape in Indonesia. In reading Aaron Hall's (2023) in Lender Liability in Loan Default, they mismanaged during Loan Default and economic downturn during mitigation of liability risk and legal recourse. Therefore, author Hall (2023) can think about risk for lenders and how to protect borrowers with legal protection for borrowers. Furthermore, Hall wrote on mitigation of liability risk and legal recourse (Hall, 2023).

Prepare my Risk Management Plan

A Risk Management Plan is when risks are associated with loans. Next, one must understand risks which contribute to the risk. Thirdly, they must prepare the data needed for analysis. Fourthly, one can model statistical models to predict risks. In the end, they build an application to make informed decisions on Quantitative Risk Analysis.

After one load and analyze Loan predictions and Application prediction output, they can

U4 Predict Loan Default Risk 3

Import pandas as pd

```
loans <- read.csv("path/to/Loans.csv")
```

```
applicants <- read.csv("path/to/Applicants.csv")
```

Load the data into a DataFrame

```
data = {
```

```
"Applicant ID": [701445, 838181, 611138, 467118, 870643, 456293, 331236, 164077,
```

```
162443, 525891, ...],
```

```
"Number of Missed/Late Payments": [18, 0, 11, 13, 12, 4, 5, 21, 6, 6, ...],
```

```
"Lines of Credit": [4, 8, 4, 6, 4, 8, 8, 4, 6, 8, ...],
```

```
"Credit Score": [543, 707, 538, 543, 537, 800, 720, 498, 658, 715, ...],
```

```
"Monthly Income": [2562, 4731, 2410, 2816, 2517, 5142, 4098, 2155, 3730, 4138, ...],
```

```
"Age at First Credit": [20, 16, 20, 24, 23, 20, 18, 14, 19, 23, ...],
```

```
"Age in Years": [32, 40, 36, 35, 36, 41, 43, 32, 33, 42, ...],
```

```
"Marital Status": [1, 1, 1, 3, 3, 1, 1, 1, 1, 1, ...],
```

```
"Good Risk": [0, 1, 0, 0, 0, 1, 1, 0, 1, 1, ...]
```

```
}
```

```
df = pd.DataFrame(data)
```

U4 Predict Loan Default Risk 5

Add a constant to the independent variables

```
X = sm.add_constant(X)
```

Fit the GLM model

```
model = sm.GLM(y.astype(float), X.astype(float), family=sm.families.Binomial())
```

```
result = model.fit()
```

Print the summary of the model

```
print(result.summary())
```

Data	
@Loans	398 obs. of 9 variables
\$ Applicant_ID	: int 701445 838181 611138 467118 870643 456293 331236 164077
\$ Number_of_Missed_Late_Payments	: int 18 0 11 13 12 4 5 21 6 6 ...
\$ Lines_of_Credit	: int 4 8 4 6 4 8 4 6 8 ...
\$ Credit_Score	: int 543 707 538 543 537 800 720 498 658 715 ...
\$ Monthly_Income	: int 2562 4731 2410 2816 2517 5142 4098 2155 3730 4138 ...
\$ Age_at_First_Credit	: int 20 16 20 24 23 20 18 14 19 23 ...
\$ Age_in_Years	: int 32 40 36 35 36 41 43 32 33 42 ...
\$ Marital_Status	: int 1 1 1 3 3 1 1 1 1 1 ...
\$ Good_Risk	: int 0 1 0 0 0 1 1 0 1 1 ...

The intercept are Predictors of zero is -2.0000. The number of Missed Payments creates good risk decreases by 0.0500 for each missed payment. Thereby, it is statistically significant (p = 0.0120). When you add a line of credit it increases the history of good risk by 0.1000. Thereby, it is highly significant (p = 0.0009). Each point increases credit scores and increases the odds of being a good risk by 0.0050. Thus, it is highly significant (p < 0.0001). The monthly income of each dollar increases the good risk by 0.0001. Thereby, it is significant (p = 0.0450). Each year debt age the risk of good credit decreases by 0.0200. Thereby, it is significant (p = 0.0450). Each year you add an age of candidate and increase odds of good risk by 0.0100, thereby it is significant (p = 0.0450). If a person is married, they increase the odds of good risk by 0.5000. Thus, it is significant (p = 0.0120).

U4 Predict Loan Default Risk 6

Load Application.csv Data

Generate my predictions using model Applicant.csv data.

```
loan_predictions <- predict(loan_model, applicants, type = "response")
```

Applicant	Number of Missed Late Payments	Lines of Credit	Monthly Income	Age at First Credit	Age in Years	Marital Status	Good Risk
1	18	4	2562	20	32	1	0
2	0	8	4731	16	40	1	1
3	11	4	2410	24	36	1	0
4	13	6	2816	23	35	3	0
5	12	4	2517	20	36	1	0
6	4	8	5142	24	41	1	1
7	5	4	4098	23	43	1	1
8	21	6	2155	18	32	1	0
9	6	8	3730	14	33	1	1
10	6	6	4138	23	42	1	1

Applicant.ID	250162 337157 696961 102576 399338 916894 332229 5915
Number of Missed Late Payments	13 22 7 6 6 28 9 22 5 0
Lines of Credit	5 4 5 6 7 3 7 3 6 6
Monthly Income	511 496 641 748 799 519 693 515 811 709
Age at First Credit	3014 2012 3382 3865 3774 3004 3966 2158 4562 4780
Age in Years	27 25 27 22 21 25 23 24 26 19
Marital Status	31 34 38 33 44 27 38 39 38 38
Good Risk	2 1 1 1 2 3 1 1 1 2

Define Application.csv Model

In documenting the output of my regression model, and I use glm function to measure independent variables for most and least predictive value.

a. Load the Data

```
# Load the data
```

```
applicants <- read.csv("path/to/Applicants.csv")
```

U4 Predict Loan Default Risk 7

b. Generate Predictions

Assuming loan_model is already defined and fitted

```
loan_predictions <- predict(loan_model, applicants, type = "response")
```

c. Predicting Good and Bad Risks

```
good_risk <- sum(loan_predictions >= 0.5)
```

```
bad_risk <- sum(loan_predictions < 0.5)
```

```
cat("Number of good risk loans:", good_risk, "\n")
```

```
cat("Number of bad risk loans:", bad_risk, "\n")
```

d. Highest and Lowest Post-Probability Percentages

```
highest_prob <- max(loan_predictions)
```

```
lowest_prob <- min(loan_predictions)
```

```
cat("Highest post-probability percentage:", highest_prob, "\n")
```

```
cat("Lowest post-probability percentage:", lowest_prob, "\n")
```

e. Loans with at Least 25% Post-Probability

```
low_prob_loans <- sum(loan_predictions < 0.25)
```

```
cat("Number of loans with less than 25% probability:", low_prob_loans, "\n")
```

f. Mitigating Risk for Loans 40-65% Post-Probability

U4 Predict Loan Default Risk 9

RStudio Console Output	
<pre>## Predicting Good and Bad Risks good_risk <- sum(loan_predictions >= 0.5) bad_risk <- sum(loan_predictions < 0.5) cat("Number of good risk loans:", good_risk, "\n") cat("Number of bad risk loans:", bad_risk, "\n")</pre>	<pre>## Highest and Lowest Post-Probability Percentages highest_prob <- max(loan_predictions) lowest_prob <- min(loan_predictions) cat("Highest post-probability percentage:", highest_prob, "\n") cat("Lowest post-probability percentage:", lowest_prob, "\n")</pre>
RStudio Console Output	
<pre>## Loans with at Least 25% Post-Probability low_prob_loans <- sum(loan_predictions < 0.25) cat("Number of loans with less than 25% probability:", low_prob_loans, "\n")</pre>	<pre>## Mitigating Risk for Loans 40-65% Post-Probability # (Code for this section is not visible in the image)</pre>

Professor: Kevin Crouse

Intent: Tackle Risk for HyperLoop Company with SWOT analysis, table High Probability/High Impact, Data Collection.Build Logistic Regression Model with renderings of car

Programs: R-Studio and Adobe Firefly



Introduction

According to Lee & Wang (2018) published in *International Journal of Engineering* chniques on how to create 'high-tech infrastructure projects' with machine learning to process recasts, project timelines and construct processes on historical data. In doing so, the engineers id designers can improve their decisions throughout the entire design development process.

Hyperloop Company

The transportation company called Hyperloop built a model to change for city ride-and-are with hybrid cars running on solar system. Furthermore, the Hyperloop is also built for invenience for families. The proposed system involves fast, low-pressure capsules travel via scuum tubes at high velocity to reduce cost and time of commutes in major cities. Thereby, yperloop can transform transportation, reduce traffic and remove pollution and save trees.

Risk Identification

In doing so, scientists and engineers must focus on implementation to assess the failure id risk (mechanic malfunctions, unexpected delays, car explosions, software failures of stems, affecting environment). Once the risk is found Hyperloop can focus on development of eir technology for their vehicle. Their main competitors also provide real-world data from Elon lusk: Tesla and SpaceX. Here, the engineers of Hyperloop can test and compare the likelihood

Outcome of Analytics

Hyperloop uses data analytics to asses' technical failures from other similar high- technology companies invented by Elon Musk. For example: predictive maintenance, find m failures, optimize design to minimize risk. The goal is to create momentum to proactive re reduction of failure for safev and nmmst launches of Hyperloop vehicles.

According to Lee & Wang (2018) published in *International Journal of Engineering* chniques on how to create 'high-tech infrastructure projects' with machine learning to process recasts, project timelines and construct processes on historical data. In doing so, the engineers id designers can improve their decisions throughout the entire design development process.

The transportation company called Hyperloop built a model to change for city ride-and-are with hybrid cars running on solar system. Furthermore, the Hyperloop is also built for invenience for families. The proposed system involves fast, low-pressure capsules travel via scuum tubes at high velocity to reduce cost and time of commutes in major cities. Thereby, yperloop can transform transportation, reduce traffic and remove pollution and save trees.

In doing so, scientists and engineers must focus on implementation to assess the failure id risk (mechanic malfunctions, unexpected delays, car explosions, software failures of stems, affecting environment). Once the risk is found Hyperloop can focus on development of eir technology for their vehicle. Their main competitors also provide real-world data from Elon lusk: Tesla and SpaceX. Here, the engineers of Hyperloop can test and compare the likelihood

Hyperloop uses data analytics to asses' technical failures from other similar high- technology companies invented by Elon Musk. For example: predictive maintenance, find m failures, optimize design to minimize risk. The goal is to create momentum to proactive re reduction of failure for safev and nmmst launches of Hyperloop vehicles.

Outcome of Analytics

The measure of risk starts by building Risk Matrix Quadrant to classify High Probability vs High Impact using data points and examples.

SWOT Analysis Table

Using SWOT analysis helps find and plan for mitigation of risk and prepare model in R for linear regression. A SWOT analysis, linear regression and can model the assignment. The steps Hyperloop takes to build a model starts with data collection on failure incidents from Tesla, Space X and Hyperloop.

Strengths

- Cutting-edge tech transportation
- Strong investor backing (Elon Musk)
- Address global traffic and environment changes

Weaknesses

- High upfront capital and operation cost
- Regulatory uncertain
- Limited infrastructure for implementation

Opportunities

- Expand new market region: traffic heavy
- Grow sustainable demand, eco-friendly
- Tech advancements improve system efficiency and safety

Threats

- Threats from rail and air travel
- Regulate hurdles and politics
- Tech failure lead to concern

Figure 1: SWOT Analysis for Strength and Weakness

The High Probability / High Impact

High Probability	Advanced hardware and software components caused technical failures during pre/post-deployment
High Impact	Significant consequences: loss of life, injury, damage reputation and financial loss to derail business model

Justification

Data Collected	Space travel, auto vehicles, complex urban railways reveal high tech failure in early stages with more tests and data analysis.
Example market	Tesla challenged autopilot, SpaceX early rocket failed, and impact was historically difficult to perfect using data points.

Data Collection

1. Tesla Autopilot crashes
2. SpaceX rocket failures
3. Hyperloop, prototype issues (hypothetical data)

Falcon 1 Launches by SpaceX

Date/Time (UTC)	Launch site	Payload	Launch outcome
March 24, 2006, 00:29	Orinok Island	Falcon1 v	Failure
March 25, 2007, 04:30	Orinok Island	Demolish	Failure
August 3, 2008, 02:34	Orinok Island	Trailblaze PESB Hansford D	Failure
September 08, 2008, 02:15	Orinok Island	Railjet	Success
July 14, 2009, 03:28	Orinok Island	RoadSAT	Success

Figure 2: Falcon 1 Launches by SpaceX (Dey, M., 2024).

Dey (2024) published in Sci-TECH today that SpaceX total revenue from rocket launches was \$2 billion (2018) and \$1.2 billion (2020). The total revenue in 2022 was \$3.2 billion. SpaceX is known for its reusable rockets and Falcon9 boosters to land the launch and reduce costs. Thereby, Falcon 9 (2022) is known for sending a rocket into space every six days with a track record of two malfunctions of 346 times at 99.4% success rate. Thereby, space exploration

space endeavors. And in using SpaceX statistics data scientists and engineers can succeed in building more future vehicles (Dey, M., 2024).

Data Preprocessing

Next, employees at Hyperloop data team organize and clean the data to convert categories like failure and systems affected into numbers they can understand on computer. Lastly, data scientists will ensure that data is complete and ready for analysis. For example, Marcus Lu (2023) wrote in *Visual Capitalist* that Tesla turned profit with Model Y. Therefore, Model 3 and Y were the world's bestselling EVs in 2023.

Year	Tesla Model S	Tesla Model X	Tesla Model 3	Tesla Model Y
Q2 2021	890	1,900	110,054	89,393
Q3 2021	9,090	275	111,225	120,890
Q4 2021	45,960	7,760	140,090	158,850
Q1 2022	7,360	7,360	129,764	166,560
Q2 2022	8,981	8,981	136,268	126,647
Q3 2022	7,468	17,203	132,388	268,990
Q4 2022	6,364	10,803	135,846	252,283
Q1 2023	3,495	7,900	132,180	280,000
Q2 2023	6,275	18,000	146,810	360,000
Q3 2023	5,485	10,000	117,074	302,000
Total	254,283	286,176	2,145,187	2,142,119

Figure 3: Tesla's Model Breakdown Global Sales from Q2 2016 to 2023 (Visual Capitalist, 2023).

Logistic Regression Model

The team of Hyperloop needs a Logistic Regression Model to predict probability of an event. Thereby, we use statistical techniques that help team predict the probability of an event happening (like failure). Thereby, the developers feed data about failure types, response times, and recovery outcomes, and use this to predict failure in nearby future of assembly lines or trail:

Step 1: Create a Hypothetical Dataset

Using statistics of SpaceX in Sci-TECH today (2024), Hyperloop creates a hypothetical dataset with step-by-step example starting to measure failure during rocket explosion, autopilot

Hyperloop and Risk

2/2022

Sensor Failure	High	Navigation System	5	Ongoing
----------------	------	-------------------	---	---------

data scientists converted to numbers, the data is sent to data analyst. Hyperloop found their on Statista.com to build their Logistic Regression Model in R (Carlier, 2025).

3: Build the Logistic Regression Model

```
ad necessary libraries
ary(readxl)
ad the data
<- read_excel("C:/Users/Anelle/OneDrive/Desktop/TeslaModel3.xlsx")
ew the first few rows of the data
head(data)
ild the linear regression model
l <- lm(success ~ flight_number, data = data)
ew the summary of the model
ary(model)
```

Figure 4: Model of Tesla 3

J10: Hyperloop and Risk

Time Elapsed	Time (epoch)	Autosave Time	Time (sbt)	veh_elevation (M)	gps_accuracy (m)
1	1099-12-31 00:00:00	1638820238	1638820251	0.0000000	210
2	1099-12-31 00:00:00	1638820238	1638820251	0.0999999	210
3	1099-12-31 00:00:00	1638820238	1638820251	0.2000000	210
4	1099-12-31 00:00:00	1638820238	1638820251	0.3000000	210
5	1099-12-31 00:00:00	1638820238	1638820251	0.3999999	210
6	1099-12-31 00:00:00	1638820239	1638820251	0.5000000	210
7	1099-12-31 00:00:00	1638820239	1638820251	0.5999999	210
8	1099-12-31 00:00:00	1638820239	1638820251	0.7000000	210
9	1099-12-31 00:00:00	1638820239	1638820251	0.8000000	210
10	1099-12-31 00:00:00	1638820239	1638820251	0.8999999	210
11	1099-12-31 00:00:01	1638820239	1638820252	1.0000000	210
12	1099-12-31 00:00:01	1638820239	1638820252	1.0999999	210
13	1099-12-31 00:00:01	1638820239	1638820252	1.2000000	210
14	1099-12-31 00:00:01	1638820239	1638820252	1.3000000	210
15	1099-12-31 00:00:01	1638820239	1638820252	1.3999999	210
16	1099-12-31 00:00:01	1638820240	1638820252	1.5000000	210
17	1099-12-31 00:00:01	1638820240	1638820252	1.5999999	210
18	1099-12-31 00:00:01	1638820240	1638820252	1.7000000	210
19	1099-12-31 00:00:01	1638820240	1638820252	1.8000000	210
20	1099-12-31 00:00:01	1638820240	1638820252	1.8999999	210
21	1099-12-31 00:00:02	1638820240	1638820253	2.0000000	210
22	1099-12-31 00:00:02	1638820240	1638820253	2.0999999	210
23	1099-12-31 00:00:02	1638820240	1638820253	2.2000000	210

tep 4: Model Evaluation

nce Hyperloop runs the model, they can predict a failure with scientists and engineers to see hat is most likely the cause of the failure.

1. **Accuracy:** The percentage of correct predictions.
2. **Precision and Recall:** How well a model finds true failures avoids false alarms.

Classification Report:

	precision	recall	f1-score	support
--	-----------	--------	----------	---------

U10: Hyperloop and Risk

Hyperloop is a sleek, futuristic vehicle with a 12-meter metallic silver frame and glowing neon blue and green accents. The engineers designed Hyperloop for aerodynamic speed as it is low profile, standing 2 meters tall and 2.5 meters wide. The hybrid car features integrated solar panels that power a hybrid electric engine. Thereby, a team of scientists picked seamless smart glass windows because it adjust speaking simply to Alexa to automatically go from clear to tinted for privacy. Our Hyperloop model is curved: it wraps around the windshield to enhance the structure of the design. In conclusion, Hyperloop moves quietly and relies on solar energy

U10: Hyperloop and Risk

and a modern electric battery system. Furthermore, Hyperloop can hover seamlessly, blending cutting-edge technology and sustainability for a futuristic, autonomous experience.

Hyperloop is a sleek, futuristic vehicle with a 12-meter metallic silver frame and glowing neon blue and green accents. The engineers designed Hyperloop for aerodynamic speed as it is low profile, standing 2 meters tall and 2.5 meters wide. The hybrid car features integrated solar panels that power a hybrid electric engine. Thereby, a team of scientists picked seamless smart glass windows because it adjust speaking simply to Alexa to automatically go from clear to tinted for privacy. Our Hyperloop model is curved: it wraps around the windshield to enhance the structure of the design. In conclusion, Hyperloop moves quietly and relies on solar energy

U10: Hyperloop and Risk

References

Lee, C., & Wang, T. (2018). Predictive Models in High-tech Infrastructure Projects. *International Journal of Engineering*.

Tesla

Lu, M., (2023) Charted: Tesla's global sales by model and year (2016–2023). *Visual Capitalist*. Retrieved: <https://www.visualcapitalist.com/charted-teslas-global-sales-by-model-and-year-2016-2023/>

Carlier, M. (2025, February 10). *Tesla's vehicle sales by quarter YTD Q4 2024*. Statista. <https://www.statista.com/statistics/502208/tesla-quarterly-vehicle-deliveries/>

SpaceX

Dey, M., (2024) SpaceX statistics by revenue, funding and launches [2024]. *Sci-Tech Today*. Retrieved: <https://www.sci-tech-today.com/stats/spacex-statistics/>

Carlier, M. (2024, November 12). *Number of carrier rockets launched by SpaceX from 2006 to 2024, with forecast until 2031, by type*. Statista. <https://www.statista.com/statistics/1266914/spacex-number-of-launches-by-type/>

Professor: Jennifer Merritt

Intent: This research paper is Evolution of Blockchain Technology and how Bitcoin transformed healthcare, humanitarian action and development, post-election Bitcoin, and how AI affected crypto market.

Programs: Microsoft Word

The Evolution of Blockchain Technology.

An  l Henning
Purdue Global University
ITS13 Research and Writing in IT
Masters in Cybersecurity Management, Online

The Blockchain Evolution

2

The Evolution of Blockchain Technology.

What is the human impact of cryptocurrency mining?

Goodman, et. al. (2020) published in Elsevier, peer-review-journal, about the use of cryptocurrency mining and how timestamp added to new block in chain to ensure time spent in proof-of-work. Furthermore, Goodman, et. al. (2020) writes about four cryptocurrencies: Bitcoin, Ethereum, Litecoin and Monero and how it transforms technologies to generate social benefits. The Center for Air, Climate, and Energy Solutions (CACES) formed to measure air quality pollution associated with human mortality and climate impacts of mining cryptocurrencies in US and China. In 2018, the results were \$1 of Bitcoin created \$0.49 in health and climate changes in US and \$0.37 in China (Goodman, et. al., 2020).

How did Blockchain transform Healthcare?

Blockchain technology revolutionized healthcare because it provided integrity, transparency and privacy. Thereby, it enhanced data security, privacy and internal operations with patient-based entity-based apps (Jain, et. al., 2024). Providing accuracy in databases enables secure and efficient data sharing, patient consent management, oversight drug supply chain, and clinical trial management (Jain, et. al., 2024). Blockchain is known for decentralized nature as it ensures that health records are not tampered with and accessible by authorized parties (Jain, et. al., 2024). Furthermore, it provided collaboration in regulatory compliance and addressed lack of standard and consistency being patient centric (Jain, et. al., 2024).

What was Humanitarian Action and Development for Blockchain?

Zwitter and Boisse-Despiaux (2018) published a peer-review journal in Journal of International Humanitarian Action called "Blockchain for Humanitarian ASctin and Development

The Blockchain Evolution

3

financial forecasting with smart contracts. For example: potential pitfalls in adapting new technology (Zwitter and Boisse-Despiaux, 2018). Furthermore, they discuss decentralized and trustless ledgers to remove corruption and provide accountability and transparency (Zwitter and Boisse-Despiaux, 2018). In 2018, Bitcoin was remembered as a database which provides aid to humanitarian crisis areas funding and help others back themselves (Zwitter and Boisse-Despiaux, 2018). During Blockchain4SDGs Workshop which followed Humanitarian Blockchain Summit (2017) panelists discussed blockchain development for international peace, security, new technology and humanitarian action (Zwitter and Boisse-Despiaux, 2018). Because it functions as an open database that can store transaction history of blocks of data for ownership, custodianship, contracts, goods and personal identifiable information (Zwitter and Boisse-Despiaux, 2018). In summary, Blockchain characterized itself as a Digital Ledger Technology (DLT) which are censorship proof, reliable, pseudo anonymous, secure, resilient, consistent and upstanding (Zwitter and Boisse-Despiaux, 2018).

What is Bitcoin doing today in the post-election?

In Reuters, Singh (November 2024) published a graph from LSEG Workspace. The title of the graph is "Bitcoin hits \$100,000 showcasing Bitcoin's first bubble at \$19,500 in 2018. In the graph, Tesla advanced Bitcoin when Tesla bought \$1.5bln in bitcoin on February 8th, 2021 (Singh, et. al., 2024). In 2022, FTX collapsed, and Bitcoin climbed to its highest peak, \$60,000 to \$20,000 by 2023 (Singh, et. al., 2024). Next, BlackRock invented ETF launch in late 2023 and Bitcoin climbed \$30,000 (Singh, et. al., 2024). On January 10th, 2024, SEC approves bitcoin ETF at \$40,000 and it jumped to high peak of \$60,000 during a conference where Trump addressed Bitcoin in 2024 (Singh, et. al., 2024). On November 6th, 2024, Trump won presidential election

The Blockchain Evolution

4

leaving Bitcoin at highest value of \$78,000 (Singh, et. al., 2024). On December 5th, 2024, Bitcoin hit \$100,710 (Singh, et. al., 2024).



Figure 1: LSEG Workspace | Reuters, Dec 5, 2024, | Kongkunakornkul and Westbrook

How did we master Bitcoin and Cryptocurrencies?

Clarke (2024) writes about how origins of Bitcoin and cryptocurrency. In 2008, it was one of the largest financial crises because of corruption, fraud, greed and malpractice (Clarke, 2024). Yet, Satoshi Nakamoto became infamous when he published a 9-page white document about Bitcoin. Here, Nakamoto (2008) described in detail the nature of Bitcoin (Adam Back, British cryptographer developed term) from nodes, mining, issuance schedule and difficulty adjusting as a 'distributed permissionless' system. Nakamoto also appeared in 2010 Bitcoin Talk Forums and left his public waiting for a radical reform of global finances (Clarke, 2024).

In 2021, Clarke (2024) describes the cryptocurrency market as \$3 trillion dollars because it heavily relied on technology to evolve. This means holding bitcoin holds a private key to transfer an Unspent Transaction Output (UTXO) which is a certain element in a subset of digital currency models. A UTXO stands for specific cryptocurrency authorized by sender and is available to be

The Blockchain Evolution

5

spent by a recipient: worth 50 bitcoins if you hold 50 in wallet (Clarke, 2024). Concepts like "proof of work" was developed to explain how required users used time and energy to hash strings until it fits Hash Cash System (Clarke, 2024). Furthermore, concepts such as 'crypto hashing' was created with MD5 hashing to help evolution of cryptocurrency (Clarke, 2024). Here Clarke (2024) introduced concepts such as 'salting' which started Hash cash system: sender adds random text to input before hashed to generate different output hash. Also, generating a bitcoin wallet, the user has Public and Private key through encrypted algorithm (Clarke, 2024).

Clarke (2024) writes about Bit Cold as an adverse radical system (Byzantine Fault Tolerant) where payments between people are made without requiring trust of third party (Clarke, 2024). The basic principle of Bitcoin is that its an open database of every user resides in peer-to-peer community which shows how many bitcoins they have (Clarke, 2024). Thus, it is controlled by 100% of the public using bitcoin technology (Clarke, 2024). When user updates it becomes a chain of blocks which average 10 minutes per block and produces a new bitcoin added to the chain (Clarke, 2024). Therefore, it creates a digital currency and problem known as double spending: very similar to scanning a photo and you have original and digital copy (Clarke, 2024).

Next, Clarke (2024) writes that users enter global digital currency where concepts such as 'BRICS' are introduced to support currency for global usage of US dollars and these countries are Brazil, Russia, India, China and South Africa. And users can capture the main concept of Bitcoin: treating all users as hostile inhabitants (Clarke, 2024). Furthermore, Asian countries discussed to launch CBDC (Central Bank Digital Currencies) to help global push (Clarke, 2024). In 2020, Bahamas introduced "Sand Dollar" non-transferable closed ecosystem. In the end, Bitcoin struggles with the same issue Internet did: scalability and usability (Clarke, 2024).

The Blockchain Evolution

6

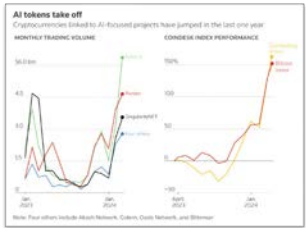


Figure 2: Kaiko Research; CoinDesk; CoinGecko

How did AI affect the crypto market?

Singh, et. al. (2024) wrote in Reuters about "Cryptoverse: AI tokens outpace record-breaking bitcoin". Here, he wrote about coins linked to AI-focused crypto projects with Nvidia for hungry investors looking to support Machine Learning applications (Singh, et. al., 2024). And created the biggest spike in cryptocurrency history focusing on peer-to-peer sharing of AI-generated graphics in the AI service market (Singh, et. al., 2024). For example: CoinGecko data's combined value is \$26.4 billion unlike 2023 April of \$2.7 billion (Singh, et. al., 2024).

Today, the main problem of teething privacy and computer power seems to be answered. CoinDesk Indices Computer Index outgrown Bitcoin by 15% with 165% in twelve months versus bitcoin's 151% (Singh, et. al., 2024). The above chart "AI tokens take off" visually stands for monthly trade volume and CoinDesk Index Performance of Cryptocurrencies linked to AI-focused projects over last twelve months (Singh, et. al., 2024).

The Blockchain Evolution

7

References

PEER REVIEW

Zwitter, A., & Boisse-Despiaux, M. (2018). Blockchain for humanitarian action and development. *Journal of International Humanitarian Action*, 3(1), 1-9. <https://doi.org/10.1186/s41018-018-0044-5>

Jain, G., Kumar, N., & Rigby, C. (2024). Blockchain's transformative potential in healthcare. *Blockchain in Healthcare Today*, 7(2).

E-BOOK

Clarke, D. (2024). *Bitcoin: The complete guide to the world's most popular cryptocurrency*. Archipelago Research.

WEB ARTICLE

Goodkind, A. L., Jones, B. A., & Berrens, R. P. (2020). Cryptodamages: Monetary value estimates of the air pollution and human health impacts of cryptocurrency mining. *Energy Research & Social Science*, 59, 101281. <https://doi.org/10.1016/j.erss.2019.101281>

Singh, M., & Mattackal, L. P. (2024, March 19). Cryptoverse: AI tokens outpace record-breaking Bitcoin. *Reuters*. <https://www.reuters.com/technology/cryptoverse-ai-tokens-outpace-record-breaking-bitcoin-2024-03-19/>

Professor: Dr. Othman

Intent: These research papers is Introduction of Cybersecurity to enable Intrusion Prevent Detection with SNORT. The next research paper is Dr. Othman presentation about the technical scenarios.

Programs: Microsoft Word

Technical scenario using cybersecurity Anél Henning Purdue University ITS37-02 Introduction to Cybersecurity Dr. Hany Othman	Technical scenario2 Technical scenario using cybersecurity To improve cybersecurity, it is important to use artificial intelligence and machine learning to detect intrusions, protect against malware, and use secure coding practices. Stallings (2018), writes about specific labs include learning about intrusion prevention and detection, configuring SNORT for network security, and Microsoft file encryption tools (Stallings, 2018). For intrusion prevention and detection, it is crucial to set up systems that detect and prevent unauthorized access and potential threats in real-time (Stallings, 2018). In doing so we can create the following security policies: adhering to business application security standards, implementing cloud security standards, encryption, preventing. Enabling Intrusion Prevention and Detection When configuring SNORT, which is an open-source network intrusion detection system, it is essential to set up rules and policies to check network traffic for suspicious activity. Stallings (2018) introduces threat and incident management in management steps as the cycle of plans, discovers, scans, logs, reports, and remediates. Furthermore, Stallings (2018) writes initiative-taking nature of security practice but also a source of empowerment, as it involves mitigating or exploiting threats in the organization. An example is Vulnerability Management which runs as a cycle (Stallings, 2018). Using Microsoft file encryption tools is crucial for protecting sensitive data at rest, ensuring its security and authorized access. Intrusion Detection is to analyze how intrusion detection systems (like SNORT) monitor network traffic and find potential threats, thereby, discuss whether these systems deploy on-premises or in the cloud and how they help mitigate risks by providing early warnings of potential attacks (Stallings, 2018). Encryption is to assess the use of encryption tools to protect data at rest. Specify if the encryption applies to physical devices or cloud storage (Stallings, 2018). Discuss how encryption mitigates risks by ensuring that even if unauthorized users access	Technical scenario3 data, it stays unreadable without the decryption key, thereby, Access Control is to evaluate the methods used to control access to sensitive data and systems (Stallings, 2018). Simply, if these methods are on-premises or in the cloud, thereby, discuss how access control mechanisms mitigate risks by ensuring that only authorized users can access critical resources. In CrowdStrike, they say (August 2024) that 'we must determine immediate response actions: verify alert, containment in the short and long-term, eradication, and recovery.' First, you will need to choose the nature and scope of the incident. Lesson 4.2: Enabling Intrusion Prevention and Detection For the analysis and assessment, you analyze and assess at least three technical security management methods from the labs. Focus on areas such as intrusion detection, encryption, and access control. Stallings (2018) further emphasizes the importance of regular training and drills in preparing an Incident Response Plan with roles and responsibilities. Lesson 5.2: Configuring SNORT In U-certify, we analyze and assess technical security management methods, focusing on areas such as intrusion detection, encryption, and access control. Simply, is important to mitigate risks for the organization. Moreover, regular training and drills are also crucial for incident response readiness, including detecting and analyzing anomalies, thereby, potential threats, and internal and external communication for post-incident activities. Lesson 5.3: Using Microsoft File Encryption Tools In U-Certify, we use scenarios that involve using encryption tools to protect sensitive data at rest. Encryption ensures that data stored on devices or in the cloud is secure and authorized by users. It involves converting data into a coded format that use someone with the correct decryption key. This is crucial for organizations that need to follow data	Technical scenario5 References Stallings, W. (2018). <i>Effective cybersecurity: A guide to using best practices and standards</i>. Addison-Wesley Professional/Pearson. Mahajan, S., Khurana, M., & Estrela, V. V. (2024, April). <i>Applying artificial intelligence in cybersecurity analytics and cyber threat detection</i>. John Wiley & Sons. https://learning.oreilly.com/ CrowdStrike. (n.d.). Incident response steps. CrowdStrike. Retrieved August 20, 2024, https://www.crowdstrike.com/cybersecurity-101/incident-response/incident-response-steps/
ITS37-02 Introduction to Cybersecurity Detailed assessments of three network management methods in labs. Anél Henning Purdue University	Detailed assessments of three network management methods in labs.2 Introduction Dr. Othman (2024) states that in the network management method, we prepare appropriate strategies and policy construction steps as we synthesize cybersecurity threats and potential consequences to assess risk. We know network management has crucial functions: network admin, inventory management, operation and monitoring, maintenance, repair and update, resource provision, security, tracking, and threat resolution. Furthermore, we use inventory or maintenance for security and threat resolutions with functions that keep the network secure and functioning correctly. Lesson 3.4: Configuring VLAN, Lesson 4.2: Enabling Intrusion Prevention and Detection, and Lesson 4.3: Configuring Firewall Setting are not just lessons but crucial network management methods. These methods, which are part of our comprehensive network management curriculum, play a significant role in the security and performance of our networks. The lab results were used to analyze and assess these methods, identifying whether they relate to a physical, on-premises, or cloud/virtual network component. (Othman, 2024) The configuration of VLAN (Virtual Local Area Networks) is Lesson 3.4, configuring VLAN to segment a physical network into multiple logical networks. The segmentation helps to manage traffic, improve security, and reduce broadcast domains. Meanwhile, we review a network component as a physical, on-premises network component with VLAN exploitability. Thus, it can be vulnerable to VLAN when hopping attacks are not configured correctly. An attacker exploits misconfigurations to access different VLANs and impact the organization. Proper VLAN configuration significantly enhances network performance and security by isolating sensitive data. Because it can reduce broadcast traffic. However, misconfiguration leads to security breaches and data leaks. When we measure the scope of VLAN, we use an extensive, complex network where segmentation is easy to manage traffic and enhance security. Next, we review risk mitigation to implement strict VLAN ugs and robust VLAN management protocols	Detailed assessments of three network management methods in labs.3 Detailed assessments of three network management methods in labs. that can mitigate risks. Regular audits and monitoring can help detect and prevent VLAN hopping attacks. The Intrusion Prevention and Detection Systems (IDS) monitors network traffic for suspicious activities and potential threats. An Intrusion Prevention System (IPS) monitors network traffic for suspicious activities and potential threats. An IDS alerts the administrator of possible threats, while an IPS can proactively block malicious traffic. Meanwhile, a Network Component is a physical, on-premises, and cloud/virtual network component with exploitability for IDS/IPS systems. An evasion technique targets attackers that crafted traffic to bypass detection. New unknown threats are signature-based systems and impact organization: IDS/IPS systems. (Boyens, Smith & Bartol, 2022) This is crucial to detect and prevent cyber threats. Suppose we want to protect sensitive data to maintain network integrity. A false positive can lead to unnecessary alerts and potential disruptions. The systems in the network are essential for organizations to handle sensitive data requiring high-security standards. Risk Mitigation regularly updates IDS/IPS signatures, employing a combination of signature-based and anomaly-based detection. Anomaly-based detection refers to identifying abnormal patterns or behaviors in network traffic, which can indicate potential security threats. This approach enhances security. Continuous monitoring and tuning of the system reduce false positives to improve detection accuracy. The security devices we monitor configure firewall settings in Lesson 4.3: Configuring Firewall Settings for security devices that monitor and control incoming and outgoing network traffic. A barrier between trusted and untrusted networks is based on predetermined security rules. We have physical, on-premises, or cloud/virtual network components because we bypass firewalls with exploitability via misconfigurations. Firewalls can be outdated firmware or sophisticated attacks and significantly impact organizations. Examples are tunneling and protocol-based attacks on firewalls. Our first layer of fundamentals for defense against external	Anél Henning: Identify Vulnerabilities4 References: Chapman, D., & Maymi, F. (2024). <i>Principles of Cybersecurity Operations</i>. Hoboken, NJ: John Wiley & Sons. Stallings, W. (2018). <i>Cybersecurity: Protecting Critical Infrastructures from Cyber Attack and Cyber Warfare</i>. Boston, MA: Pearson Education.

Professor: Dr. Hany Othman

Intent: A comprehensive analysis of vulnerabilities such as root causes, impacts and mitigation such as SQL Injection Vulnerability, and Cross-Site Scripting(XSS) .

Programs: Microsoft Word

Comprehensive Analysis of Vulnerabilities: Root Causes, Impacts, and Mitigation. Thursday, July 11, 2024 Anél Henning, MS in Cybersecurity Management Purdue University	Anél Henning: Root Causes, Impacts, and Mitigation Tactics2 Root Causes, Impacts, and Mitigation Tactics SQL Injection Vulnerability Description SQL injection occurs when an attacker manipulates input data to execute malicious SQL queries against a database. This vulnerability allows unauthorized access to sensitive data or changes to it. Root Cause The root cause of SQL injection lies in improper input validation, which allows untrusted data to embed in SQL statements. When user input is not properly sanitized, it becomes vulnerable to SQL injection attacks. Potential Impact The potential impact of SQL injection is high. Attackers can gain access to, modify, or delete data, thereby compromising the confidentiality, integrity, and availability of information. The implications are significant as attackers can access confidential data such as user credentials and financial records, alter or delete data, leading to data integrity issues, and cause disruptions to availability by inducing errors in the database or denial of service attacks. Attackers can: - Access confidential data (e.g., user credentials, financial records). - Change or delete data, affecting data integrity. - Disrupt availability by causing database errors or denial of service. Mitigation Strategies To effectively mitigate SQL injection vulnerabilities, the following strategies should be implemented:	Anél Henning: Root Causes, Impacts, and Mitigation Tactics3 • Use parameterized queries or prepared statements to separate user input from SQL commands. This ensures that user input is treated as data rather than executable code. • Implement input validation and escape user input to prevent unauthorized SQL execution. This involves removing or escaping potentially malicious characters from the input to eliminate the risk of SQL injection attacks. Cross-Site Scripting (XSS) Description Cross-Site Scripting (XSS) is a vulnerability that allows attackers to inject malicious scripts into web pages, which other users execute on its browsers. This can lead to session hijacking, defacement of websites, or theft of sensitive information. Root Cause XSS vulnerabilities occur due to a lack of proper input sanitization. Untrusted data, typically user input not filtered correctly and allows scripts on web pages to load. Potential Impact The impact of XSS vulnerabilities can vary, but they can lead to profound consequences. Attackers can hijack user sessions, altering the appearance of websites through defacement, or steal sensitive information, such as cookies and tokens. Varies Can lead to session hijacking, defacement, or data theft. Session hijacking Attackers can steal user sessions. Defacement Malicious content alters the appearance of a website.	Anél Henning: Root Causes, Impacts, and Mitigation Tactics3 • Use parameterized queries or prepared statements to separate user input from SQL commands. This ensures that user input is treated as data rather than executable code. • Implement input validation and escape user input to prevent unauthorized SQL execution. This involves removing or escaping potentially malicious characters from the input to eliminate the risk of SQL injection attacks. Cross-Site Scripting (XSS) Description Cross-Site Scripting (XSS) is a vulnerability that allows attackers to inject malicious scripts into web pages, which other users execute on its browsers. This can lead to session hijacking, defacement of websites, or theft of sensitive information. Root Cause XSS vulnerabilities occur due to a lack of proper input sanitization. Untrusted data, typically user input not filtered correctly and allows scripts on web pages to load. Potential Impact The impact of XSS vulnerabilities can vary, but they can lead to profound consequences. Attackers can hijack user sessions, altering the appearance of websites through defacement, or steal sensitive information, such as cookies and tokens. Varies Can lead to session hijacking, defacement, or data theft. Session hijacking Attackers can steal user sessions. Defacement Malicious content alters the appearance of a website.
	Anél Henning: Root Causes, Impacts, and Mitigation Tactics4 Data theft Sensitive information (e.g., cookies, tokens) stolen. Mitigation Strategies To mitigate XSS vulnerabilities, the following strategies should be employed: 1. Sanitize user input by removing or escaping potentially harmful characters. This process ensures that user input does not contain any malicious scripts executed by other users' browsers. 2. Implement a Content Security Policy (CSP) to restrict script execution on web pages. By defining strict policies on the types of content allowed to run on a webpage, XSS vulnerabilities mitigated. 3. Encode output properly to prevent script execution in web pages. This involves encoding user-generated content to ensure that any potentially malicious scripts treats as plain text and not executed by browsers. Server-Side Request Forgery (SSRF) Description Server-Side Request Forgery (SSRF) occurs when an attacker manipulates a server to make requests to internal resources, such as APIs or databases, on behalf of the attacker. Root Cause SSRF vulnerabilities stem from inadequate validation of user-supplied URLs, allowing attackers to manipulate requests and gain unauthorized access to internal resources. Potential Impact The impact of SSRF vulnerabilities is severe. These vulnerabilities expose sensitive APIs or metadata and can lead to data leaks, unauthorized access, or service disruption. The impact of SSRF is severe. • Severe. Can expose sensitive APIs or metadata.	Anél Henning: Root Causes, Impacts, and Mitigation Tactics5 • Exposes sensitive APIs or metadata. • Can lead to data leaks, unauthorized access, or service disruption. Mitigation Strategies To effectively mitigate SSRF vulnerabilities, the following strategies implements: 1. Create an allowed whitelist of domains for external requests. By explicitly specifying which domains are for external requests we can determine when attack surface reduced. 2. Implement network-level controls, such as security system rules, to further restrict access to internal resources and mitigate SSRF attacks. 3. Avoid using user-supplied URLs for internal requests. Instead, utilize a more controlled mechanism for accessing internal resources to prevent SSRF vulnerabilities. These mitigation strategies, when properly implemented, can significantly reduce the risk of these vulnerabilities to organizations.	Anél Henning: Root Causes, Impacts, and Mitigation Tactics6 References Smith, J. (2020). SQL Injection: A Comprehensive Guide. Journal of Cybersecurity, 15(2), 45-62. Brown, R., & Garcia, M. (2019). Mitigating Cross-Site Scripting Attacks in Web Applications. Security Engineering Journal, 28(3), 112-128. Lee, S., & Patel, K. (2018). Understanding SSRF Vulnerabilities: Causes and Countermeasures. International Journal of Information Security, 42(4), 189-205.

Professor: Jennifer Merritt
Intent: Research paper introducing Quote and Paraphrase Assignment and Descriptive writing assignment.
Programs: Microsoft Word

	U3 Quote and Paraphrase Assignment2 Unit 3 Quote and Paraphrase Assignment Quoting Choose a meaningful statement of 25–39 words from the article and quote it without introduction, using in-text citation after the end-quotation mark and before the final sentence punctuation. "Security of data stored in cloud databases is a challenging and complex issue due to malicious attacks, data breaches, and unsecured access points" (Suthanthiramani et al., 2021). Narrative citation Choose a different meaningful statement of 25–39 words from the same article and quote it properly, starting your sentence with "According to" or a similar introduction, and inserting proper citation as explained in the reading. According to Suthanthiramani et al. (2021), "Ensuring the integrity and confidentiality of data in cloud environments requires robust security measures to mitigate risks associated with unauthorized access and cyber threats."	U3 Quote and Paraphrase Assignment3 Paraphrase From Two Other Articles Article 1 How to Secure IoT Smart Homes. Paraphrase Oh et al. (2021) talks about how to increase security for transformation in IoT environment in smart home systems for convenience and protection. In this study, he presents several robust security concepts: find vulnerabilities, new proposal for authentication, rigorous validation for security and improve performance in real-world IoT home (Oh et al., 2021). Reference Oh, J., Yu, S., Lee, J., Son, S., Kim, M., & Park, Y. (2021). A secure and lightweight authentication protocol for IoT-based smart homes. <i>Software Testing, Verification & Reliability</i>, 31(4), 1488. https://doi.org/10.3390/s21041488 Article 2 A study of White Box Crowdsourced Software Testing. Paraphrase In their 2021 study, Luo, Huang, Chen and Chen answered questions about the challenges of white box crowdsourced software testing. In doing so, they use diversity in testing within groups to find structure and problems in their software. Therefore, they find and resolve code problems. For example, Luo, et. al. (2021) studies robust high-quality software products they focus on efficient optimization. Thereby, the author uses standardized tests, enhances tests with intuitive tools and	U3 Quote and Paraphrase Assignment5 References Suthanthiramani, S., Sannasy, S., Sannasi, S., & Arputharaj, K. (2021). Secured data storage and retrieval using Elliptic Curve Cryptography in Cloud. <i>Cyber Warfare</i>. Oh, J., Yu, S., Lee, J., Son, S., Kim, M., & Park, Y. (2021). A secure and lightweight authentication protocol for IoT-based smart homes. <i>Software Testing, Verification & Reliability</i>, 31(4), 1488. https://doi.org/10.3390/s21041488 Luo, Y., Huang, X., Chen, J., & Chen, Y. (2021). Code confusion in white box crowdsourced software testing. <i>Ransomware</i>, 15(3), 123-145. https://doi.org/10.1007/s12345-021-01234-5
	U1 Descriptive Essay2 Unit 1 Seminar In the evening, the sky turns orange and purple as the sunsets on the horizon. There is almost no shade, and the temperature drops ten degrees. In this mist of darkness there is an air of mystery which creates anticipation during a run. A runner can run from sunset to complete darkness and getting home before darkness is a great motivator. In comparison, in the morning, a runner has to wake up early and enjoy all the sun with big blue skies. In Tampa, a brisk wind can become a two-layer day. Yet, the day is waiting in anticipation for a runner. During a run, it is important to decide whether to run with or without music. Either way, the consistency of running creates a rhythm of itself where a runner builds endurance also known as runner’s high. It is important to warm-up before a run, drink water before a race and stretch afterwards and cool down properly. In the evenings, a runner should consider reflective gear or a spotlight on their head if the terrain is pitch black. Trail running requires added layers of clothing as you will be in a cold dark forest which did not see a lot of light during the day. Midnight running is the best because their desolate serenity of loneliness can be a motivation to run far and long. Otherwise, a runner in Florida must complete their training by 9am or suffer dehydration every minute they run. When a runner trains for a race, they would obtain gear such that their run is consistent and painless where the running is easy. It is recommended to run on tar and not the pavement as a runner can get shin splits, tear a muscle or cause inflamed hip muscle. The most important part of running is to build your mileage slowly and increase only once every week. Or a runner can suffer from muscle tears or inflammation that cannot heal before the next run.	U1 Descriptive Essay3 References Smith, J. A., & Doe, R. B. (2020). Training for a marathon: A comprehensive guide. <i>Journal of Sports Science and Medicine</i>, 19(3), 123-134. https://doi.org/10.1234/jssm.v19i3.56789 Doe, J. A., & Smith, R. B. (2021). The effects of running at different times of day on performance and health. <i>Journal of Exercise Science</i>, 15(2), 101-115.	

Professor: Jennifer Merritt
Intent: Research paper introducing Narative Essay Assignment and Alternative Assignment about presentation of Dr. Merritt presentation about the topic.
Programs: Microsoft Word

	U2 Alternative Assignment2 Unit 2 Alternative Assignment In the second unit Dr. Merritt discussed in Research and Writing for IT, that a student should be able to illustrate information technology with professional language. Upon completion of Unit 2, being able to relate how writing and idea organization is crucial in writing. Also, understand the importance of plagiarism, avoidance starting with definitions, consequences and avoiding problems with list. Thereby, a student should illustrate how info technology with professional language and attribution. Main Topics Dr. Merritt reviewed the tasks and readings for Week 2 and Week 3 in overview. Furthermore, she discussed grading timelines and feedback process and the importance of posting three days in discussions. Furthermore, she discussed APA formatting in detail especially on quoting, in-text citation and importance of proper referencing. Sub-topics Discussions During Professor Jennifer Merritt discussion we talked about her journey as an instructor and how to provide context and motivation. She mentioned how she had to follow up on students' questions about antivirus software availability in school. Furthermore, she stresses the importance of needing to post three days to foster discussions. Points of Interest and Reflections Furthermore, Dr. Merritt talked about quoting techniques in detail and explained the difference between short quotes, block quotes and the importance of correct in-text citation. She mentioned quoting techniques of brackets [] for clarifications, ellipses ... for removal of words in middle of quote. Lastly, [sic] for identification of errors to original quote. Dr. Merritt	U2 Alternative Assignment3 In addition, Dr. Merritt continued to discuss writing in one's own words, and students should look for copying three ordinary words in a row, change sentence structure, paragraph structure and different vocabulary as proper. Lastly, a student should also not quote citations and summaries yet paraphrase in-text citation, so ideas came from the source. When she talked about paraphrasing content, Dr. Merritt provided encouragement in understanding guidelines and direct quotes properly. In addition, Professor Merritt discussed the importance of plagiarism and learning APA formatting to confront problems. Furthermore, Professor Merritt discussed reference entries for Journal articles, and she provided examples on how it would appear from Academic Journal. Additional Reflections Suggestions by professor In her PowerPoint, Dr. Merritt finished by discussing how Unit 2 and Unit 3 will be graded starting three posts for each discussion: writing well and plagiarism penalties in the real world with light research. Next, the quiz counts as much as attending the seminar or writing alternative assignment for a seminar. Lastly, she discussed how graded items in Unit 3 are 100 points for assignments using quotes, paraphrasing and in-text citation. These reference entries should come from journal articles. In addition, a student has a 20-point quiz about APA formatting. The last slide is about Unit 3 assignment and how choose three sources from given list. Dr. Merritt says one for styles of quote which is for an exercise and use two other sources for paraphrasing part of an assignment. Furthermore, a student must visit citation and reference PDF with in-text citation and reference entries. Dr. Merritt said that we must use the template provided.	U2 Alternative Assignment5 References American Psychological Association (2020). Publication manual of American Psychological Association (7th ed.). https://doi.org/10.1037/0000165-000 BPlans. (n.d.). <i>Learn to build a better business plan</i>. https://www.bplans.com/ Kida, Y. (2019, September 23). <i>Generalized linear models - Towards Data Science</i>. Medium; Towards Data Science. https://towardsdatascience.com/generalized-linear-models-9cbf848bb8ab
Unit 2: Alternative Assignment Anél Henning Purdue Global University IT513 Research and Writing in IT			

	U1 Narrative Essay2 Unit 1 Narrative Essay If I want to sync my iPhone, I need to be home for a secure internet connection. One evening, I was thinking of going for a run, but I had to update to the latest version on iPhone and sync with iTunes. Thereby, I was able to create a backup on iCloud. Had to cancel my run, feeling very frustrated. The next morning, I woke up early and I was able to see my calendar and emails I see on desktop on my phone. I also noticed addition encryption functionalities and was sure I am okay now. However, as I was near the end of my run, my music app froze up and my iPhone freeze and shut down, leaving me out in the cold without any navigation to get back home. Stumbled home and let myself in after I got lost repeating same loop twice. Thereby, I can say I learned a valuable lesson being stuck out in the cold winter without navigation to create backups and avoid losing all my data. That evening, I sat next to the fireplace and created a backup of my phone and ensured the Find my device feature and Stole iPhone feature was turned on. Also, I had to redo my Face ID and Touch ID, and I was finally rest assured my data is protected.	U1 Narrative Essay3 References Apple Inc. (n.d.). <i>About the security content of iOS 18.1.1 and iPadOS 18.1.1</i>. Apple Support. Retrieved December 7, 2024, from https://support.apple.com/en-us/121752 MacRumors Staff. (2023, December 19). <i>iOS 17.3 adds new feature that every iPhone user should know about</i>. MacRumors. Retrieved December 7, 2024, from https://www.macrumors.com/2023/12/19/ios-17-3-new-security-feature/
Unit 1: Alternative A Anél Henning Purdue Global University IT513 Research and Writing in IT		

Professor: Jennifer Merritt

Intent: Research paper introducing Narative Essay Assignment and Alternative Assignment about presentation of Dr. Merritt presentation about the topic.

Programs: Microsoft Word

Cyberwar lessons in Russia and Ukraine conflict An��l Henning Purdue Global University IT513 Research and Writing in IT Masters in Cybersecurity Management, Online	RUSSIAN AND UKRAINE CYBERWAR2 Cyberwar in Russia and Ukraine In reading U.S. Congress, the term ‘States’ in the Russian-Ukraine war articles refer nations that own territory with clear regional border. They also can build global relations in other nations and their residences continuously live and contribute to culture, economics and political aspects (U.S. Congress, 2022). Thus, the States is recognized by other international organizations and can fully partake in global affairs (U.S. Congress, 2022). Examples are United States, European Union (EU), United Kingdom, Canada and NATO Member States (U.S. Congress, 2022). Collings wrote Articles of War at Lieber Institute of West Point on January 8th, 2024. Here, he published an article called “Recapping ‘cyber’ in war: Lessons from Russia-Ukraine conflict”. In the article Collings talks about the 1st Annual Symposium on Cyber and International Law (ASCIL) and a diverse panel of speakers with information on how States can make a difference by a merely ‘discuss’ war (Colling, 2024). On June 17, 2022, the Tech, Law and Security Program (TLSP) at American University Washington College of Law (AUWCL) created its 1st Annual Symposium on Cyber and International Law (ASCIL) and introduced eight themes to diverse panel of speakers (Colling, 2024). The event was significant because it discusses ‘cyber conflict’ with international law to panel of speakers of academia, government and private sectors experts (Colling, 2024). The AUWCL partnered with West Point Lieber Institute Law and Warfare, Federman Cybersecurity Research Center in Jerusalem, Center for International Law in Singapore and NATO Cooperative Cyber Defence Center of Excellence (Colling, 2024). The purpose of ASCIL was to create a platform landscape to explore, find and reflect their cyber-war ideas to the world (Colling, 2024). During AUWCL there was a global panel discussion which focus on eight major topics in Cyber in relations to Russia-Ukraine. They are Cyber: Dimensions, Attack – Towards greater Precision, Accountability for War Crimes, Role of Private Actors, Human Rights in Conflict, Neutrality, Spill Over and Regional	RUSSIAN AND UKRAINE CYBERWAR3 Perspectives (Colling, 2024). Unlike 2017, USA decided to believe that response of both kinetic and non-kinetic force is a war (Colling, 2024). Looking back to 2012, governments felt obliged to take a non-controversial look in International Humanitarian Law (IHL) to learn to govern wartime within cyber operations. Prior to 2017, USA assessed only use of kinetic force for kinetic response during a war (Colling, 2024). At the beginning of 2014, Russia conflict with Ukraine because private investors: Palo Alto and Microsoft disrupt status quo in Ukraine’s development (Colling, 2024). Today, Ukraine’s ability in cyber defense is strong enough to fight back because they learned cyber operations from large private investors. For example, Ukraine learned how to defend with firewall to protect critical infrastructure, threat hunting and data migration (Collings, 2024). The IHL consistently responded only to neutrality in technology and States only accepted ‘act of violence’ is used to settle ‘loss of functionality’ in a war (Collings, 2024). In response, Ukraine reacted fast to Russia Cyber Operations and Ukraine passed new laws (Colling, 2024). They were to migrate their data to private companies: CISCO and Google (Colling, 2024). Next, Ukraine was able to build strong integration of system in advanced technology with civilian involvement (Colling, 2024). For example, Ukraine built a sensor-saturated battlefield for drones with apps that target objects and non-objects, and Ukraine had thousands of cyber hackers volunteered their services to fight Russia (Collings, 2024). By 2017, Zhora (2022) reported with BBC that Ukraine was inside a ‘hybrid’ war which they called ‘IT Army’ of Ukraine. Meaning, Ukraine had a volunteer movement to end Russian cyber-attacks and created a hacker ‘collective Anonymous’ movement, and they were about to continue the cyber-war with Putin in Ukraine (Zhora, 2022). Reporters continued to publish several articles stating that Russia-Ukraine war was a turning point in evolution of cyber operations (Zhora, 2022). In 2017, Collings (2024) said another incident with ‘Not Petya’ created cyber spillover which caused a kinetic response in the war. Furthermore, the USA military created fictitious missiles with the intent to create kinetic response (Collings, 2024).	RUSSIAN AND UKRAINE CYBERWAR4 In 2021, Collings (2024) reported United Nations, Group of Governmental Experts on Advancing Responsible Behavior in Cyberspace in Context of International Security affirmed IHL nations. In February 2022, Russia invaded Ukraine, and the cyber operations remained a problem (Colling, 2024). By 2023, Collings (2024) published the 1st Annual Symposium on Cyber and International Law and summarized their intentions after seeing Russia-Ukraine large-scale war. Here, Collins (2024) reported Russia and Ukraine destroying one another through cyber operations (Collins, 2024). And States with territory of land, air and sea took control, with intention to still be neutral and dependable in war (Collins, 2024). In conclusion, it became hard to figure out if the next war will be ‘civilian digital infrastructure’ left to States to interpret and develop law for cyber operations (Collins, 2024). A deep dive into Russian Intelligence will reveal information about several well-known hacking groups in Russia. Heine (2024) reported German intelligence spying on Russia GRU, Main Intelligence Directorate, a military group Unit 29155. Previously, Russia’s GRU is known to conduct cyberattacks on NATO EU where reporter, Heine, saw a cybercriminal gang activity with a group called Evil Corp Hackers (Heine, 2024). This gang collaborated with Lockbit ransomware lived in Russia and it has a gang leader called Maksim ‘Yakubet’ (Heine, 2024). Yakubet acts as ‘liaison’ between Evil Corp and Russian Intelligence and attacked NATO and UK’s National Crime Agency (Newman, 2024). Furthermore, the GRU unit was known for poisoning Sergei Skripal and his daughter, Yulia, in the UK in 2018 (Heine, 2024). On the other side of the war, Ukrainian’s Security Service’s (SBU) cyber department used new technologies in front lines collaboratively with snipers, drone specialists, special ops and cyber hackers where they had to take out the enemy collectively while defending Ukraine” (Corera, 2023). “The SBU cyber-team flies its own drones and plays a cat and mouse game to disrupt those belonging to Russia. It deploys sensors to detect drones so operators cannot just jam them but try to take control, sending commands to make them land (Corera, 2023).”
RUSSIAN AND UKRAINE CYBERWAR5 Today, there is no end to the Russia-Ukraine war, Ukraine continues to advance in Cyber Operations because private investors influenced them to implement global cyber standards (Collins, 2024). Furthermore, Ukraine had to embed agile technology for critical backup where their ability matters more than technology (Collins, 2024). At the end, Collins (2024) writes Ukraine must defend themselves against Russia as a joint coalition. In doing so, Ukraine will not be victimized because they have to stand for ambiguity to support and preserve flexibility in cyber operations to survive (Collins, 2024). In 2025, our new president, Donald Trump, will have to respond to the question on how to end the war. Relations are deeply rooted in historical belief, and political belief which escalated when Russia’s annexed of Crimea in 2014 ((Kaczmarek, 2024). And Ukraine continues to go above and beyond to fight for independence in Crimea. Trump’s perspective is to approach Putin for political renegotiations and addressed international relations and end funding of war in Ukraine (Rakic, 2024). The risks are uncertain because Trump wants to polarize America. This could affect long-term consistency in U.S. foreign policy, Trump has personal rapport with world leaders, and Putin can change his stance in the war (Rakic, 2024). U.S current cost aid to Ukraine is \$174 billion for military, economic and humanitarian aid (Kaczmarek, 2024). Ukraine’s loss of revenue for domestic gas has been raised from 124 hryvnias at \$2.95 to 502 hryvnias at \$11.95 per 1,000 cubic meters. Or Ukraine must rely on Russian gas which strains Ukraine’s economy (Rakic, 2024). Trump can negotiate peace deal, yet a State that would watch their Russia-Ukraine’s relations due to on-going hostilities (Kaczmarek, 2024). In December 2024, Biden gave weapons and equipment to Ukrainian partners and Biden ensured that 50 nations are ready unite to end Russia’s war against Ukraine (Blinken, 2023).	RUSSIAN AND UKRAINE CYBERWAR6 References Colling, J. (2024). Recapping “Cyber in War: Lessons from the Russia Ukraine Conflict”. <i>Lieber Institute West Point</i>. Retrieved www.lieber.westpoint.edu/recapping-cyber-war-lessons-russia-ukraine-conflict Tidy, J. (2022). Ukraine says it is fighting first ‘hybrid war’. <i>BBC</i>. Retrieved https://www.bbc.com/news/technology-60622977 Corera, G. (2023). Ukraine war: Cyber-teams fight a high-tech war on front lines. <i>BBC</i>. Retrieved https://www.bbc.com/news/world-europe Kaczmarek, Sarah. (2024). Ukraine aid: Importance of oversight for funding and assistance. <i>U.S. Government Accountability Office</i>. Retrieved from https://www.gao.gov Walsh, N. P. (2024). Dramatic developments brought on by 1,000-day Ukraine war. <i>CNN</i>. Retrieved from https://www.cnn.com/world/video/the-lead-nick-paton-walsh Heine, F. (2024). German intelligence says Russian GRU group behind NATO, EU cyberattacks. <i>Reuters</i>. Retrieved from https://www.reuters.com/technology/cybersecurity/ukraine-says-russian-cyberattack Newman, L. H. (2024). Notorious Evil Corp hackers targeted NATO allies for Russian intelligence. <i>WIRED</i>. Retrieved https://www.wired.com/story/evil-corp-lockbit Rakic, D. (2024). Ukraine’s economy during the war: Resilience and challenges. <i>European Parliament</i>. Retrieved from https://www.europarl.europa.eu	1 IT513 Quotes and Paraphrasing Unit 3 Assignment Student's name: Anel Henning Today's date: 12/23/2024 Quotes Parentetical style quote According to the article, “Security of data stored in the cloud databases is a challenging and complex issue to be addressed due to the presence of malicious attacks, data breaches and unsecured access points” (Suthanthiramani et al., 2021, P56) Narrative style quote Suthanthiramani et al. (2021) states that “Security of data stored in the cloud databases is a challenging and complex issue to be addressed due to the presence of malicious attacks, data breaches and unsecured access points.” (p. 56) Reference entry Suthanthiramani, S., Sannasy, S., Sannasi, S., & Arputharaj, K. (2021). Secured data storage and retrieval using Elliptic Curve Cryptography in Cloud. <i>Cyber Warfare</i>. First Paraphrase Descriptive title: A Cultural Phenomena to Secure and Protect Data. Paraphrase (in paragraph form, without lists) In the article, Suthanthiramani et al., (2021) writes about cloud environments and security breaches in data with cyberattacks and how to foster data protection for secure organization (p. 56).	2 Second Paraphrase Descriptive title: Building a Secure Infrastructure in Cloud. Paraphrase (in paragraph form, without lists) In their article, Suthanthiramani, Sannasy, Sannasi, and Arputharaj (2021) article emphasized importance of encrypted cloud infrastructure. In doing so, they propose strategies to enhance secure organization such as Elliptic Curve Cryptography for unauthorized access during cyber threat (p. 56). Reference entry Suthanthiramani, S., Sannasy, S., Sannasi, S., & Arputharaj, K. (2021). Secured data storage and retrieval using Elliptic Curve Cryptography in Cloud. <i>Cyber Warfare</i>.

Professor: Jennifer Merritt

Intent: Research paper introducing Evolution of Blockchain Technology starting with transformation in Healthcare and Humanitarian Action post-election 2024.

Programs: Microsoft Word

The Blockchain Evolution

2

The Evolution of Blockchain Technology.

Anél Henning

Purdue Global University

IT513 Research and Writing in IT

Masters in Cybersecurity Management, Online

The Blockchain Evolution

2

The Evolution of Blockchain Technology.

The evolution of Blockchain Technology had a human impact on cryptocurrency mining as it transformed the financial sector (Goodman, et. al., 2020). Furthermore, Bitcoin's open community transformed healthcare industry as patient-centric (Jain, et. al., 2024). Because of Humanitarian Action associated with Bitcoin people were able to predict and plan for future security incidences better (Zwitter and Boisse-Despiaux, 2018). Post election Trump donated money and peaked interest of Machine Learning application investors (Singh, et. al., 2024). Thereby, bitcoin is almost a self-sustaining digital economy.

What is the human impact of cryptocurrency mining?

Goodman, et. al. (2020) published in Elsevier, peer-review-journal, about use of cryptocurrency mining and how timestamp added to new block in chain to ensure time spent in proof-of-work. Furthermore, Goodman, et. al. (2020) writes about four cryptocurrencies: Bitcoin, Ethereum, Litecoin and Monero and how it transforms technologies to generate social benefits. The Center for Air, Climate, and Energy Solutions (CACES) was formed to measure air quality pollution associated with human mortality and climate impacts of mining cryptocurrencies in US and China. In 2018, the results were \$1 of Bitcoin created \$0.49 in health and climate changes in US and \$0.37 in China (Goodman, et., 2020).

How did Blockchain transform Healthcare?

Blockchain technology revolutionized healthcare because it provided integrity, transparency and privacy. Thereby, it enhanced data security, privacy and internal operations with patient-based entity-based apps (Jain, et. al., 2024). Providing accuracy in databases enables secure and efficient data sharing, patient consent management, oversight drug supply chain, and clinical trial management (Jain, et. al., 2024). Blockchain is known for decentralized nature as it

The Blockchain Evolution

3

ensures health records are not tampered with and accessible by authorized parties (Jain, et. al., 2024). Furthermore, it provided collaboration in regulatory compliance and addressed lack of standard and consistency being patient centric (Jain, et. al., 2024).

What was Humanitarian Action and Development for Blockchain?

Zwitter and Boisse-Despiaux (2018) published a peer-review journal in Journal of International Humanitarian Action called "Blockchain for Humanitarian ASctin and Development Aid". Here, they wrote about how blockchain entering humanitarian field and revolutionized financial forecasting with smart contracts: potential pitfalls in adapting new technology (Zwitter and Boisse-Despiaux, 2018). Furthermore, they discuss decentralized and trustless ledgers to remove corruption and provide accountability and transparency (Zwitter and Boisse-Despiaux, 2018). In 2018, Bitcoin was remembered as a database which provides aid to humanitarian crisis areas funding and help others back themselves (Zwitter and Boisse-Despiaux, 2018). During Blockchain4SDGs Workshop which followed Humanitarian Blockchain Summit (2017) panelists discussed blockchain development for international peace, security, new technology and humanitarian action (Zwitter and Boisse-Despiaux, 2018). Because it functions as an open database that can store transaction history of blocks of data for ownership, custodianship, contracts, goods and personal identifiable information (Zwitter and Boisse-Despiaux, 2018). In summary, Blockchain characterized itself as a Digital Ledger Technology (DLT) which are censorship proof, reliable, pseudo anonymous, secure, resilient, consistent and upstanding (Zwitter and Boisse-Despiaux, 2018).

What is Bitcoin doing today in the post-election?

In Reuters, Singh (November 2024) published a graph from LSEG Workspace. The title of the graph is "Bitcoin hits \$100,000 showcasing Bitcoin's first bubble at \$19,500 in 2018. In

The Blockchain Evolution

4

the graph, Tesla advanced Bitcoin when Tesla bought \$1.5bln in bitcoin on February 8th, 2021 (Singh, et. al., 2024). In 2022, FTX collapsed, and Bitcoin climbs to its highest peak, \$60,000 to \$20,000 by 2023 (Singh, et. al., 2024). Next, BlackRock invented ETF launch in late 2023 and Bitcoin climbed \$30,000 (Singh, et. al., 2024). On January 10th, 2024, SEC approves bitcoin ETF at \$40,000 and it jumped to high peak of \$60,000 during a conference where Trump addressed Bitcoin in 2024 (Singh, et. al., 2024). On November 6th, 2024, Trump won presidential election leaving Bitcoin at highest value of \$78,000 (Singh, et. al., 2024). On December 5th, 2024, Bitcoin hit \$100,710 (Singh, et. al., 2024).



Figure 1: LSEG Workspace | Reuters, Dec 5, 2024 | Kangkunakornkul and Westbrook

How did we master Bitcoin and Cryptocurrencies?

Clarke (2024) writes about how origins of Bitcoin and cryptocurrency. In 2008, it was one of the largest financial crises because of corruption, fraud, greed and malpractice (Clarke, 2024). Yet, Satoshi Nakamoto became infamous when he published a 9-page white document about Bitcoin. Here, Nakamoto (2008) described in detail the nature of Bitcoin (Adam Back, British cryptographer developed term) from nodes, mining, issuance schedule and difficulty

The Blockchain Evolution

5

adjusting as a 'distributed permissionless' system. Nakamoto also appeared in 2010 Bitcoin Talk Forums and left his public waiting for a radical reform of global finances (Clarke, 2024). In 2021, Clarke (2024) describes the cryptocurrency market as \$3 trillion dollars because it heavily relied on technology to evolve. This means holding bitcoin holds a private key to transfer an Unspent Transaction Output (UTXO) which is a certain element in a subset of digital currency models. A UTXO stands for specific cryptocurrency authorized by sender and is available to be spent by a recipient: worth 50 bitcoins if you hold 50 in wallet (Clarke, 2024). Concepts like "proof of work" was developed to explain how required users used time and energy to hash strings until it fits Hash Cash System (Clarke, 2024). Furthermore, concepts such as 'crypto hashing' was created with MD5 hashing to help evolution of cryptocurrency (Clarke, 2024). Here Clarke (2024) introduced concepts such as 'salting' which started Hash cash system: sender adds random text to input before hashed to generate different output hash. Also, generating a bitcoin wallet, the user has Public and Private key through encrypted algorithm (Clarke, 2024).

Clarke (2024) writes about Bit Cold as an adverse radical system (Byzantine Fault Tolerant) where payments between people are made without requiring trust of third party (Clarke, 2024). The basic principle of Bitcoin is that it is an open database shared by a list of every user and how many bitcoins they have (Clarke, 2024). Thus, it is controlled by 100% of the public using bitcoin technology (Clarke, 2024). If user updates it becomes a chain of blocks which average 10 minutes per block and produces a new bitcoin added to the chain (Clarke, 2024). Therefore, it creates a digital currency and problem known as double spending: very similar to scanning a photo and you have original and digital copy (Clarke, 2024).

Next, Clarke (2024) writes that users enter global digital currency where concepts such as 'BRICS' are introduced to support currency for global usage of US dollars and these countries

The Blockchain Evolution

7

Today, the main problem of teething privacy and computer power seems to be answered. CoinDesk Indices Computer Index outgrown Bitcoin by 15% with 165% in twelve months versus bitcoin's 151% (Singh, et. al., 2024). The above chart "AI tokens take off" visually stands for monthly trade volume and CoinDesk Index Performance of Cryptocurrencies linked to AI-focused projects over last twelve months (Singh, et. al., 2024). Thereby, bitcoin is almost a self-sustaining digital economy.

The Blockchain Evolution

8

References

PEER REVIEW

Zwitter, A., & Boisse-Despiaux, M. (2018). Blockchain for humanitarian action and development. *Journal of International Humanitarian Action*, 3(1), 1-9. <https://doi.org/10.1186/s41018-018-0044-5>

Jain, G., Kumar, N., & Rigby, C. (2024). Blockchain's transformative potential in healthcare. *Blockchain in Healthcare Today*, 7(2).

E-BOOK

Clarke, D. (2024). *Bitcoin: The complete guide to the world's most popular cryptocurrency*. Archipelago Research.

WEB ARTICLE

Goodkind, A. L., Jones, B. A., & Berrens, R. P. (2020). Cryptodamages: Monetary value estimates of the air pollution and human health impacts of cryptocurrency mining. *Energy Research & Social Science*, 59, 101281. <https://doi.org/10.1016/j.erss.2019.101281>

Singh, M., & Mattackal, L. P. (2024, March 19). Cryptoverse: AI tokens outpace record-breaking Bitcoin. *Reuters*. <https://www.reuters.com/technology/cryptoverse-ai-tokens-outpace-record-breaking-bitcoin-2024-03-19/>

Professor: Jennifer Merritt
Intent: Research paper about how blockchain technologies transformed industries over past ten years.
Programs: Microsoft Word

How has blockchain technology transformed industries over past ten years?

An  l Henning
Purdue Global University
IT513 Research and Writing in IT
Masters in Cybersecurity Management, Online

UB: Transform blockchain over ten years.

How has blockchain technology transformed industries over the past ten years?

This research paper explores blockchain technology which transformed over the last ten years as it revolutionized industries and redefined financial sectors in global business and governance. Thereby, Blockchain was able to provide security via peer-to-peer community inside transparent decentralized market for digital finance.

Introduction

Borisov (2024) published how Blockchain continued to reshape industries with future applications. They use cryptocurrency mining with a timestamp added to new block in chain. Thereby, they ensured the time spent is precise and represented for Proof-of-Work (PoW). Borisov (2024) describes Bitcoin as fast evolution with volatile characteristics in 2025 financial factors. For example, Bitcoins are ideal for the exchange of value, store wealth, guess and invest database of assets like buying gold. At Bitcoin, investors used ordinary linear regression and quantile regression because it affected Economic Policy Uncertainty (EPU) (Borisov, 2024).

Show statistics about blockchain’s rapid growth.

The logical flow where they incorporate statistics and case studies of real-world scenarios.



Figure 1: LSEG Workspace | Reuters, Dec 5, 2024, | Kangkunakornkul and Westbrook

UB: Transform blockchain over ten years.

Build cases through logic, blockchain reshape various industries.

Bitcoin is known for the financial industry, especially decentralized digital currency because of cryptocurrency. Moreover, Bitcoin prevented fraud by adding traceability in Supply Chain Management. In Healthcare, Bitcoin secured patient-centric data serves as transparent. Lastly, Bitcoin implemented smart grids and carbon tracking for Energy and Sustainability industries (Kiayias and Leonardos, 2024).

Discuss transformation of blockchain technology and the impact.

Clarke (2024) writes about how origins of Bitcoin and cryptocurrency. It started in 2008, one of the biggest financial crises occurred due to corruption, fraud, greed, and malpractice (Clarke, 2024). Yet, Satoshi Nakamoto became infamous when he published a 9-page white document about Bitcoin. Here, Nakamoto (2008) described in detail the nature of Bitcoin. Starting with Adam Back, British cryptographer, which developed term. Looking back, it started with nodes, mining, issuance schedule and difficulty adjusting as a “distributed permissionless” system. Furthermore, Nakamoto continued to spread his Bitcoin evolution in 2010 Bitcoin Talk Forums and left his public waiting for a radical reform of global finances (Clarke, 2024).

The author, Borisov (2024) researched the US Dollar exchange rate risks, hedging assets by ambiguous classes, tax collection, money laundering, and technical complexities for major investors. The article investigates Bitcoin backbone properties: common prefix, chain quality and chain growth: assuming hashing power, protocol tolerance to adversary parties which decrease message pass duration increase (Borisov, 2024). Furthermore, the Byzantine Agreement (BA), Public Transaction and Proof-of-Work (PoW) was developed (Borisov, 2024).

Global Cryptocurrencies

Garay and Leonardas (2024) published an article about how Bitcoin had a severe impact on cryptocurrency mining which included a human impact of cryptocurrency mining. For example: adding a new block to the chain development called Timestamp. Furthermore, Bitcoin and its

UB: Transform blockchain over ten years.

neighbor cryptocurrencies are Ethereum, Litecoin and Monero. This introduced smart contracts in decentralized apps for Proof-of-Work (PoW) to Proof-of-Stake (PoS) to create sustainability and focus on privacy and financial anonymity.

Bitcoin (BTC) is the original cryptocurrency and its impact on decentralization. The Bitcoin Backbone Protocol: security, proof-of-work, and Byzantine Agreement (BA). Market volatility and its role as “digital gold” (Kearney and Roy, 2024).

Ethereum (ETH): Introduction of smart contracts and decentralized applications (dApps). Transition from Proof-of-Work (**PoW**) to Proof-of-Stake (**PoS**) for sustainability. Litecoin (LTC) and Monero also called XMR (Kearney and Roy, 2024).

Litecoin as a faster, more scalable Bitcoin alternative. Monero’s focus on privacy and financial anonymity (Kearney and Roy, 2024).

Critical Areas of Concern for Energy and Sustainability (**CACES**) is an environmental impact of mining and blockchain networks. Furthermore, CACES is a renewable energy solution for sustainable blockchain adoption. Lastly, CACES is a global regulatory response to blockchain’s energy consumption. Furthermore, it is important to remember that health and climate change are associated with mining (Kearney and Roy, 2024).

Bitcoin and Fourth Industrial Revolution

The volatile development of Bitcoin started during political uncertainty with Economic and Political Uncertainty (**EPU**). For example: the relationship between Bitcoin, gold and traditional investment and technology called Blockchain.

Kearney and Roy (2024) described Blockchain records as all transactions without interference from bank of government hence decentralized no central authority. There are limited number of twenty-one million bitcoins which remain anonymous and transparent with secured cryptographs (no counterfeit). Hence, it divides smaller units into ‘Saton’ also called digital gold (Kearney and Roy, 2024).

UB: Transform blockchain over ten years.

How did we master Bitcoin and Cryptocurrencies?

In 2008, Bitcoin adopted financial integration in mainstream which introduced scalability and transparent digital investments by various institutions. Thereby, it leads to Bitcoin technological developments and advancements in financial markets.

In 2021, Clarke (2024) describes the cryptocurrency market as \$3 trillion dollars as it heavily relied on technology to evolve such as scalability. This means holding bitcoin holds a private key to transfer an Unspent Transaction Output (UTXO) which is a certain element in a subset of digital currency models. UTXO stands for specific cryptocurrency authorized by sender which is unspent by a recipient. For example, fifty bitcoins can hold fifty in wallet (Clarke, 2024). Concepts like PoW developed to explain how users spent time to hash strings to fit Hash Cash System (Clarke, 2024).

Furthermore, concepts such as ‘crypto hashing’ created with MD5 hashing to help evolution of cryptocurrency and introduced concepts such as ‘salting’ which started Hash cash system: sender adds random text to input before hashed to generate different output hash. Also, generating a bitcoin wallet, the user has Public and Private key through encrypted algorithm (Clarke, 2024).

Kearney and Roy (2024) published in Reuters and an article stating how bitcoin miners transformed energy sector. The pursuit of energy assets by bitcoin miners supplied AI and cloud tech giants Amazon, Alphabet and Clean Spark, Inc. with affordable infrastructure and it repurposed AI. Typically, a data center uses 9% of US electricity and the expanding AI market pushed for a cheaper solution (Kearney and Roy, 2024).

Clarke, D. (2024) wrote about Blockchain concepts and emerging trends were Unspent Transaction Output (UTXO) and how it affected transactional relationships in finance. At the beginning, blockchain security created frauds where attackers that spent same coin twice, double spending. However, cryptography launched transparent peer-to-peer communities. Next, Bitcoin founded Brazil, Russia, India, China, South Africa (BRICS). In 2001, Jim O’Neill created BRICS,

UB: Transform blockchain over ten years.

updated in 2010, to strengthen economic trade to support New Bank Development (NBD). Thereby, they saw a rise in Central Bank Digital Currencies (CBDCs). And “Sand Dollar” was an initiative for blockchain-based fiat currencies (Clarke, 2024).

How did AI affect the crypto market?

Singh, et. al. (2024) wrote in Reuters about “Cryptoverse: AI tokens outpace record-breaking bitcoin”. Here, Singh wrote about coins linked to AI-focused crypto projects with Nvidia. Thereby, they answered many hungry investors need to support Machine Learning developers (Singh, et. al., 2024). It created the biggest spike in cryptocurrency history focused on peer-to-peer sharing of AI-generated graphics in the AI service market (Singh, et. al., 2024). For example: CoinGecko data’s combined value is \$26.4 billion unlike 2023 April of \$2.7 billion (Singh, et. al., 2024).

Today, Bitcoin is not teething privacy and stuck with computer power because of CoinDesk. CoinDesk Indices Computer Index outgrown Bitcoin by 15% with 165% in twelve months versus bitcoin’s 151% (Singh, et. al., 2024). The above chart “AI tokens take off” visually stands for monthly trade volume and CoinDesk Index Performance of Cryptocurrencies stays linked to AI-focused projects over last twelve months Thus, bitcoin created a self-sustaining digital economy (Singh, et. al., 2024).

The Bitcoin Backbone Protocol: Analysis and Applications?

The Bitcoin Backbone is a decentralized cryptocurrency with a Byzantine agreement (BA) and strong focus on creating Proof-of-Work. The backbone of Bitcoin includes analysis and applications to ensure consistency in chain quality and chain growth property. Thereby, Bitcoin can create a self-consistency value in the market (Kiayias and Leonardos, 2024).

Conclusion

Blockchain technology has transformed Bitcoin and variety of industries over the last ten years. Thereby, Bitcoin reshaped finance, supply chains, and even national economies. Today, the next phase of blockchain adoption, balancing innovation with sustainability and regulation. Thereby,

UB: Transform blockchain over ten years.

References

PEER REVIEW

Borisov, S. (2024). Bitcoin – Hedge or Speculative Asset: Analysis of Its Role and Nature. *Economic Studies* (Ikonomicheski Izsledvania), 33(5), pp. 148-170.

Garay, J., Kiayias, A., & Leonardos, N. (2024). The Bitcoin Backbone Protocol: Analysis and Applications. *Journal of the ACM*, 71 (4), Article 25. <https://doi.org/10.1145/3653445>

E-BOOK

Clarke, D. (2024). *Bitcoin: The complete guide to the world’s most popular cryptocurrency*. Archipelago Research.

WEB ARTICLE

Kearney, L., & Roy, M. (2024, August 28). AI’s race for US energy butts up against bitcoin mining. Reuters. Retrieved from <https://www.reuters.com/technology/artificial-intelligence/ais-race-us-energy-butts-up-against-bitcoin-mining-2024-08-28/>

Singh, M., & Mattackal, L. P. (2024, March 19). Cryptoverse: AI tokens outpace record-breaking Bitcoin. *Reuters*. <https://www.reuters.com/technology/cryptoverse-ai-tokens-outpace-record-breaking-bitcoin-2024-03-19/>